TEXTBOOK OF AGRICULTURAL STATISTICS · CHAPTER 1

5-MINUTE REVISION

Basics of Statistics

From the chapter's own revision points, formulae and questions.

Selected from Chapter 1 · Chapter Summary in the textbook.

Textbook authors: **Dr. Pratheesh P. Gopinath · Dr. Adarsh V.S. · Mohammed Hisham M. · Dr. Manju Mary Paul**

STAT 3202 · Chapter revision

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Quick revision · 1

- A population is the complete group of interest; a sample is a part selected for study.
- A parameter describes a population, while a statistic is calculated from sample observations.
- The measurement scale determines which numerical comparisons are meaningful.
- Class intervals must cover the data without overlap; boundaries depend on the recording precision.
- Frequencies sum to the number of observations, and relative frequencies sum to one.

Quick revision · 2

- The source, selection method and reference period must be checked before interpreting data.

Important formulae · 1

Number of classes using Sturges' formula:

$$k = 1 + 3.322 \log_{10} N$$

Approximate class width:

$$C = \frac{|\max - \min|}{k}$$

A possible centring rule for a preliminary lower endpoint:

$$L = \min - \frac{C'k' - (\max - \min)}{2}$$

This is an optional starting value, not a necessary rule. Final limits and boundaries must respect the measurement precision, cover the data and avoid overlapping classes.

Important formulae · 2

Lower class boundary of the ith class:

$$L_i = \frac{\text{Lower class limit of } i + \text{Upper class limit of } (i-1)}{2}, \quad i = 2, 3, 4, \ldots$$

Upper class boundary of the ith class:

$$U_i = \frac{\text{Upper class limit of } i + \text{Lower class limit of } (i+1)}{2}, \quad i = 1, 2, 3, \ldots$$

Class mark:

$$\text{Class mark} = \frac{\text{Lower class limit} + \text{Upper class limit}}{2}$$

Important formulae · 3

Class width:

$$C = \text{Upper class boundary} - \text{Lower class boundary}$$

Relative frequency:

$$\text{Relative frequency} = \frac{f}{N}$$

Sum of frequencies:

$$\sum f = N$$

Important formulae · 4

Sum of relative frequencies:

$$\sum \frac{f}{N} = 1$$

Self-check from the chapter

1. The branch of statistics concerned with summarizing data is called _____ statistics. **[Descriptive]**
2. The branch of statistics concerned with drawing conclusions about a population from a sample is called _____ statistics. **[Inferential]**
3. A numerical constant describing a population is termed a _____. **[Parameter]**
4. A subset of a population selected for study is called a _____. **[Sample]**

TEXTBOOK OF AGRICULTURAL STATISTICS · CHAPTER 2

5-MINUTE REVISION

Official Agricultural Statistics

From the chapter's own revision points, formulae and questions.

Selected from Chapter 2 · Chapter Summary in the textbook.

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Quick revision · 1

- Crop production combines an area estimate and a corresponding yield estimate.
- CCE plots must be selected by the prescribed sampling procedure, not by convenience.
- Plot yield requires the net produce weight and the actual harvested area.
- Villages, fields and experimental plots can form successive sampling stages.
- Official estimates should be quoted with their reference year and release status.

Quick revision · 2

- The Agricultural Census describes operational holdings.

Important formulae

Diagonal of a square CCE plot:

$$AC = BD = \sqrt{5^2 + 5^2} = 7.07 \text{ m}$$

Diagonal of a square of side a:

$$d = a\sqrt{2}$$

Self-check from the chapter

1. Crop production is estimated by multiplying crop area by average crop _____. **[Yield]**
2. The field procedure used to estimate crop yield is called a crop _____ experiment. **[Cutting]**
3. The full form of CCE is _____. **[Crop Cutting Experiment]**
4. The full form of GCES is _____. **[General Crop Estimation Survey]**

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5-MINUTE REVISION

Graphical Representation

From the chapter's own revision points, formulae and questions.

Selected from Chapter 3 · Chapter Summary in the textbook.

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Quick revision · 1

- Histogram → continuous quantitative data; adjacent bars.
- Bar chart → categorical data; bars separated by gaps.
- Equal class width → height represents frequency.
- Unequal class width → height represents frequency density.
- Less than ogive → upper class boundaries + LCF.

Quick revision · 2

- Greater than ogive → lower class boundaries + GCF.
- Intersection of two ogives → median.
- Frequency polygon → class midpoints + frequencies.
- Stem-and-leaf plot → retains individual observations.
- Simple bar chart → one value for each category.

Quick revision · 3

- Component bar chart → shows components of each category.
- Grouped bar chart → compares multiple values within categories.
- Pie chart → sectors represent parts of a whole.
- Boxplot → five-number summary and spread of data.
- IQR → $Q_3 - Q_1$.

Quick revision · 4

- Outliers in Boxplot → observations below $Q_1 - 1.5IQR$ or above $Q_3 + 1.5IQR$.

Important formulae · 1

Frequency density:

$$\text{Frequency Density} = \frac{\text{Class Frequency}}{\text{Class Width}}$$

Angle of a sector in a pie chart:

$$\text{Sector angle} = \frac{\text{Value of category}}{\text{Sum of category values}} \times 360^\circ$$

Interquartile range:

$$IQR = Q_3 - Q_1$$

Important formulae · 2

Lower limit for identifying outliers in a boxplot:

$$Q_1 - 1.5 \times IQR$$

Upper limit for identifying outliers in a boxplot:

$$Q_3 + 1.5 \times IQR$$

Self-check from the chapter

1. A graph representing a continuous frequency distribution by adjoining rectangles is called a _____. **[Histogram]**
2. For unequal class widths, histogram heights should represent frequency _____. **[Density]**
3. A graph of cumulative frequencies is called an _____. **[Ogive]**
4. The intersection of the less-than and greater-than ogives provides an estimate of the _____. **[Median]**

5-MINUTE REVISION

Central Tendency I

From the chapter's own revision points, formulae and questions.

Selected from Chapter 4 · Chapter Summary in the textbook.

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Quick revision · 1

- Mean → arithmetic average of observations.
- Median → middle value after arranging observations.
- Mode → most frequently occurring value.
- Mean → uses all observations.
- Median → based on position.

Quick revision · 2

- Mode → based on frequency.
- Mean is affected by extreme values.
- Median is less affected by extreme values.
- Mode is useful for identifying the most common category or value.
- Mean + constant → mean increases by the same constant.

Quick revision · 3

- Mean × constant → mean is multiplied by the same constant.
- Adding a constant to every observation does not change the standard deviation.
- Multiplying every observation by a constant multiplies the standard deviation by the absolute value of that constant.
- Sum of deviations from the arithmetic mean = 0.
- Standard deviation = 0 → all observations are identical.

Quick revision · 4

- For positive observations, $AM \geq GM \geq HM$.
- Symmetric unimodal distribution with a finite mean and a central mode → mean = median = mode.
- In many familiar positively skewed unimodal distributions, mean > median > mode; this is not a universal theorem.
- In many familiar negatively skewed unimodal distributions, mean < median < mode; this is not a universal theorem.

Important formulae · 1

Arithmetic mean:

$$\bar{x} = \frac{\sum x_i}{n}$$

Arithmetic mean for a frequency distribution:

$$\bar{x} = \frac{\sum f_i x_i}{\sum f_i}$$

Assumed mean method:

$$\bar{x} = A + \frac{\sum f_i d_i}{\sum f_i}$$

where

$$d_i = x_i - A$$

Important formulae · 2

Coding method:

$$\bar{x} = A + C\frac{\sum f_i u_i}{\sum f_i}$$

where

$$u_i = \frac{x_i - A}{C}$$

Weighted mean:

$$\bar{x}_w = \frac{\sum w_i x_i}{\sum w_i}$$

Median for odd n:

$$Md = \left(\frac{n+1}{2}\right)^{\text{th}} \text{observation}$$

Important formulae · 3

Median for even n:

$$Md = \frac{1}{2}\left[\left(\frac{n}{2}\right)^{\text{th}} \text{observation} + \left(\frac{n}{2}+1\right)^{\text{th}} \text{observation}\right]$$

Median for grouped data:

$$Md = L + \left(\frac{\frac{N}{2} - F}{f}\right)C$$

Mode for grouped data:

$$Mo = L + \left(\frac{f_m - f_p}{2f_m - f_p - f_s}\right)C$$

Important formulae · 4

Relationship among arithmetic mean, geometric mean, and harmonic mean:

$$AM \geq GM \geq HM$$

Worked example from the chapter

Find the mean of the numbers 2, 4, 7, 8, 11, 12

Solution

$$\overline{x} = \frac{2+4+7+8+11+12}{6} = \frac{44}{6} = 7.33$$

Self-check from the chapter

1. The sum of observations divided by their number is called the arithmetic _____. **[Mean]**
2. The sum of deviations from the arithmetic mean is _____. **[Zero]**
3. The average calculated using assigned weights is called the _____ mean. **[Weighted]**
4. The middle value of an ordered dataset is called the _____. **[Median]**

5-MINUTE REVISION

Central Tendency II

From the chapter's own revision points, formulae and questions.

Selected from Chapter 5 · Chapter Summary in the textbook.

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Quick revision · 1

- Geometric mean → suitable for growth rates, ratios, percentages, and multiplicative changes.
- Harmonic mean → suitable for rates, ratios, and reciprocal relationships.
- Geometric mean → nth root of the product of n observations.
- Harmonic mean → reciprocal of the arithmetic mean of reciprocals.
- Geometric mean gives greater weight to smaller values and less weight to larger values.

Quick revision · 2

- Harmonic mean gives greater weight to smaller values.
- The logarithmic GM requires positive observations; the product GM of non-negative values is zero if any value is zero.
- The standard statistical GM is generally unsuitable for negative observations.
- Arithmetic mean, geometric mean, and harmonic mean are called Pythagorean means.
- Mathematical averages → arithmetic mean, geometric mean, and harmonic mean.

Quick revision · 3

- Positional measures → median, quartiles, percentiles, and deciles; mode is based on frequency.
- Quartiles → divide data into 4 equal parts.
- Q_1 → 25th percentile.
- Q_2 → 50th percentile → Median.
- Q_3 → 75th percentile.

Quick revision · 4

- Percentiles → divide data into 100 equal parts.
- $P_{25} = Q_1$.
- $P_{50} = Q_2 = \text{Median}$.
- $P_{75} = Q_3$.
- Deciles → divide data into 10 equal parts.

Quick revision · 5

- $D_5 = Q_2 = P_{50} = \text{Median}$.
- For raw data, arrange observations in ascending order before finding positional measures.
- For discrete frequency data, use cumulative frequencies to locate the required position.
- For grouped frequency data, first identify the relevant quartile, percentile, or decile class.
- In grouped data, interpolation is then used to calculate the required value.

Important formulae · 1

Geometric mean:

$$GM = \sqrt[n]{\prod_{i=1}^{n} x_i}$$

Geometric mean using logarithms:

$$\log GM = \frac{\sum_{i=1}^{n} \log x_i}{n}$$

Geometric mean using antilogarithms:

$$GM = \text{Antilog}\left(\frac{\sum_{i=1}^{n} \log x_i}{n}\right)$$

Important formulae · 2

Geometric mean for a frequency distribution:

$$GM = \text{Antilog}\left(\frac{\sum_{i=1}^{k} f_i \log x_i}{n}\right)$$

Harmonic mean:

$$HM = \frac{n}{\sum_{i=1}^{n} \frac{1}{x_i}}$$

Harmonic mean for a frequency distribution:

$$HM = \frac{n}{\sum_{i=1}^{k} f_i \frac{1}{x_i}}$$

Important formulae · 3

First quartile for raw data:

$$Q_1 = \left(\frac{n+1}{4}\right)^{\text{th}} \text{ item}$$

Third quartile for raw data:

$$Q_3 = \left(\frac{3(n+1)}{4}\right)^{\text{th}} \text{ item}$$

First quartile for grouped data:

$$Q_1 = l_1 + \frac{\frac{n}{4} - m_1}{f_1} \times c_1$$

Important formulae · 4

Third quartile for grouped data:

$$Q_3 = l_3 + \frac{\frac{3n}{4} - m_3}{f_3} \times c_3$$

Percentile for raw data:

$$P_p = \left(\frac{p(n+1)}{100}\right)^{\text{th}} \text{ item}$$

Percentile for grouped data:

$$P_p = l + \frac{\left(\frac{pn}{100}\right) - cf}{f} \times c$$

Important formulae · 5

Decile for raw data:

$$D_d = \left(\frac{d(n+1)}{10}\right)^{\text{th}} \text{ item}$$

Decile for grouped data:

$$D_d = l + \frac{\left(\frac{dn}{10}\right) - cf}{f} \times c$$

Relationship among Pythagorean means (the first equality is for two positive observations):

$$AM \times HM = GM^2$$

$$AM \geq GM \geq HM$$

Worked example from the chapter

A car travels a certain distance from City A to City B at a speed of 60 km/h, and returns the same distance from City B to City A at a speed of 90 km/h. What is the average speed for the entire trip?

Solution

To find the average speed when traveling the same distance at two different speeds, we use the harmonic mean. The harmonic mean for the speed is $\frac{2 \cdot S_1 \cdot S_2}{S_1 + S_2}$, where $S_1 = 60\,\text{km/h}$ is the speed from City A to City B, $S_2 = 90\,\text{km/h}$ is the speed from City B to City A.

In this case, we are calculating the average speed over the entire round trip. The harmonic mean is used because it accounts for the fact that traveling at different speeds over the same distance results in an overall average speed that is closer to the lower of the two speeds, rather than simply averaging the two speeds.

Now, applying the harmonic mean formula:

$S_{\text{avg}} = \frac{2 \cdot 60 \cdot 90}{60+90} = \frac{10800}{150} = 72\,\text{km/h}$

So, the average speed for the entire trip is 72 km/h.

Self-check from the chapter

1. For positive observations, the nth root of their product is the _____ mean. **[Geometric]**
2. The reciprocal of the arithmetic mean of reciprocals is the _____ mean. **[Harmonic]**
3. The mean suitable for averaging positive growth factors is the _____ mean. **[Geometric]**
4. The mean suitable for averaging speeds over equal distances is the _____ mean. **[Harmonic]**

5-MINUTE REVISION

Measures of Dispersion

From the chapter's own revision points, formulae and questions.

Selected from Chapter 6 · Chapter Summary in the textbook.

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Quick revision · 1

- Dispersion → extent to which observations deviate from a central value.
- Range → difference between largest and smallest observations.
- Range → simplest measure of dispersion.
- IQR → spread of the middle 50% of observations.
- IQR → $Q_3 - Q_1$.

Quick revision · 2

- IQR is less affected by extreme values than range.
- MAD → average absolute deviation from a central value.
- MAD about the median is the least.
- Variance → average squared deviation, using $n - 1$ for sample variance.
- Standard deviation → non-negative square root of sample variance.

Quick revision · 3

- Variance is expressed in squared units.
- Standard deviation is expressed in the original units.
- Standard deviation is always non-negative.
- Standard deviation = 0 → all observations are identical.
- Larger standard deviation → greater dispersion, when datasets are measured on the same scale.

Quick revision · 4

- Standard deviation is an absolute measure of dispersion.
- Coefficient of variation is a relative measure of dispersion.
- Higher CV → greater relative variability.
- Lower CV → lower relative variability.
- CV is useful for comparing datasets measured in different units or scales.

Quick revision · 5

- Range uses only the two extreme observations.
- IQR uses the middle 50% of the observations.
- MAD uses all observations but ignores the signs of deviations.
- Standard deviation uses all observations and gives greater weight to extreme deviations because deviations are squared.

Important formulae · 1

Range:

$$R = L - S$$

where L is the largest observation and S is the smallest observation.

Interquartile range:

$$IQR = Q_3 - Q_1$$

Mean absolute deviation:

$$MAD = \frac{\sum_{i=1}^{n} |x_i - \bar{x}|}{n}$$

Important formulae · 2

Sample variance:

$$s^2 = \frac{\sum_{i=1}^{n} (x_i - \bar{x})^2}{n - 1}$$

Sample standard deviation:

$$s = \sqrt{\frac{\sum_{i=1}^{n} (x_i - \bar{x})^2}{n - 1}}$$

Alternative formula for sample variance:

$$s^2 = \frac{1}{n-1} \left\{ \sum_{i=1}^{n} x_i^2 - \frac{1}{n} \left(\sum_{i=1}^{n} x_i \right)^2 \right\}$$

Important formulae · 3

Variance for a discrete frequency distribution:

$$s^2 = \frac{1}{n-1}\left\{\sum_{i=1}^{k} f_i x_i^2 - \frac{1}{n}\left(\sum_{i=1}^{k} f_i x_i\right)^2\right\}$$

Standard deviation for a discrete frequency distribution:

$$s = \sqrt{\frac{1}{n-1}\left\{\sum_{i=1}^{k} f_i x_i^2 - \frac{1}{n}\left(\sum_{i=1}^{k} f_i x_i\right)^2\right\}}$$

For grouped frequency data:

$$d_i = \frac{x_i - A}{c}$$

where x_i is the class mark, A is the class mark with the highest frequency, and c is the class interval.

Sample variance for grouped frequency data:

$$s^2 = \frac{1}{n-1}\left\{\sum_{i=1}^{k} f_i d_i^2 - \frac{1}{n}\left(\sum_{i=1}^{k} f_i d_i\right)^2\right\} c^2$$

Important formulae · 4

Sample standard deviation for grouped frequency data:

$$s = c\sqrt{\frac{1}{n-1}\left\{\sum_{i=1}^{k} f_i d_i^2 - \frac{1}{n}\left(\sum_{i=1}^{k} f_i d_i\right)^2\right\}}$$

Coefficient of variation:

$$CV = \frac{s}{\bar{x}} \times 100$$

Worked example from the chapter

Consider the measurement on yield and plant height of a paddy variety. The mean and standard deviation for yield are 50 kg and 10 kg respectively. The mean and standard deviation for plant height are 55 cm and 5 cm respectively. Compare the variability.

Solution

Here, the measurements for yield and plant height are in different units. Hence the variability can be compared only by using coefficient of variation.

For yield, CV = $\frac{10}{50} \times 100 = 20\%$

For plant height, CV = $\frac{5}{55} \times 100 = 9.1\%$

The yield is subject to more variation than the plant height.

Self-check from the chapter

1. The spread of observations is called _____. **[Dispersion]**
2. The difference between the largest and smallest observations is the _____. **[Range]**
3. The difference between the third and first quartiles is the _____ range. **[Interquartile]**
4. Half the interquartile range is called the quartile _____. **[Deviation]**

5-MINUTE REVISION

Skewness and Kurtosis

From the chapter's own revision points, formulae and questions.

Selected from Chapter 7 · Chapter Summary in the textbook.

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Quick revision · 1

- Skewness → degree of asymmetry of a distribution.
- Kurtosis → fourth standardised central moment, associated with tail behaviour and extreme observations.
- For a symmetric unimodal distribution with an existing mean and a unique central mode, mean = median = mode.
- Symmetric distribution → skewness = 0.
- Negatively skewed → longer left tail.

Quick revision · 2

- Many unimodal left-skewed distributions have mean < median < mode; this is not universal.
- Positively skewed → longer right tail.
- Many unimodal right-skewed distributions have mean > median > mode; this is not universal.
- Karl Pearson's measure → based on mean, mode or median, and standard deviation.
- Bowley's measure → based on quartiles.

Quick revision · 3

- Kelley's measure → based on percentiles, particularly P_{10}, P_{50}, and P_{90}.
- Moment-based skewness → based on the third central moment.
- β_1 → indicates the extent of skewness but not its direction.
- γ_1 → indicates both magnitude and direction of skewness.
- $\gamma_1 = 0$ → zero third-moment skewness; symmetry is not established.

Quick revision · 4

- $\gamma_1 > 0$ → positively skewed.
- $\gamma_1 < 0$ → negatively skewed.
- $\beta_2 = 3$ → mesokurtic.
- $\beta_2 > 3$ → leptokurtic.
- $\beta_2 < 3$ → platykurtic.

Quick revision · 5

- $\gamma_2 = 0$ → mesokurtic.
- $\gamma_2 > 0$ → leptokurtic.
- $\gamma_2 < 0$ → platykurtic.
- $\mu_1 = 0$ for every distribution.
- μ_2 is the population variance.

Quick revision · 6

- μ_3 is used for measuring skewness.
- μ_4 is used for measuring kurtosis.

Important formulae · 1

Karl Pearson's coefficient of skewness:

$$S_k = \frac{\text{Mean} - \text{Mode}}{\text{Standard deviation}}$$

Alternative Karl Pearson's coefficient of skewness:

$$S_k = \frac{3(\text{Mean} - \text{Median})}{\text{Standard deviation}}$$

Bowley's coefficient of skewness:

$$S_Q = \frac{(Q_3 - Q_2) - (Q_2 - Q_1)}{(Q_3 - Q_2) + (Q_2 - Q_1)}$$

Important formulae · 2

Alternative form of Bowley's coefficient:

$$S_Q = \frac{Q_3 - 2Q_2 + Q_1}{Q_3 - Q_1}$$

Kelley's coefficient of skewness:

$$S_P = \frac{(P_{90} - P_{50}) - (P_{50} - P_{10})}{(P_{90} - P_{50}) + (P_{50} - P_{10})}$$

rth central moment:

$$\mu_r = \frac{\sum_{i=1}^{k} f_i (x_i - \bar{x})^r}{N}$$

Important formulae · 3

Zeroth central moment:

$$\mu_0 = 1$$

First central moment:

$$\mu_1 = 0$$

Second central moment:

$$\mu_2 = \sigma^2$$

Important formulae · 4

Third central moment:

$$\mu_3 = \frac{\sum_{i=1}^{k} f_i (x_i - \bar{x})^3}{N}$$

Measure of skewness based on moments:

$$\beta_1 = \frac{\mu_3^2}{\mu_2^3}$$

Karl Pearson's γ_1:

$$\gamma_1 = \frac{\mu_3}{\mu_2^{3/2}} = \text{sgn}(\mu_3)\sqrt{\beta_1}$$

Important formulae · 5

Measure of kurtosis:

$$\beta_2 = \frac{\mu_4}{\mu_2^2}$$

Kurtosis of excess:

$$\gamma_2 = \beta_2 - 3$$

Self-check from the chapter

1. The asymmetry of a distribution is measured by _____. **[Skewness]**
2. A distribution with a longer right tail is _____ skewed. **[Positively]**
3. A distribution with a longer left tail is _____ skewed. **[Negatively]**
4. Bowley's coefficient of skewness is based on _____. **[Quartiles]**

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5-MINUTE REVISION

Measures of Association

From the chapter's own revision points, formulae and questions.

Selected from Chapter 8 · Chapter Summary in the textbook.

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Quick revision · 1

- Measures of association → describe relationships between variables.
- Linear relationship → relationship that can be represented by a straight line.
- Monotonic relationship → one variable consistently increases or consistently decreases as the other increases, but not necessarily at a constant rate.
- Scatter diagram → graphical method for examining association between two variables.
- Positive correlation → both variables tend to move in the same direction.

Quick revision · 2

- Negative correlation → variables tend to move in opposite directions.
- No correlation → no systematic linear association.
- Simple correlation → relationship between two variables.
- Multiple correlation → relationship involving more than two variables.
- Linear correlation → relationship represented by a straight line.

Quick revision · 3

- Non-linear correlation → relationship represented by a curved pattern.
- Partial correlation → relationship between two variables after controlling other variables.
- Perfect correlation → all points lie on a straight line.
- Pearson's correlation → measures linear association.
- Pearson's r ranges from -1 to $+1$.

Quick revision · 4

- $r = +1$ → perfect positive correlation.
- $r = -1$ → perfect negative correlation.
- $r = 0$ → no linear correlation.
- Covariance → measures joint linear variability.
- Covariance depends on the units of measurement.

Quick revision · 5

- Pearson's correlation is unit-free.
- Pearson's correlation is sensitive to outliers.
- Pearson's correlation summarises linear association; a nonlinear pattern can make it misleading as a general measure of association.
- Spearman's ρ → rank-based measure of monotonic association.
- Spearman's correlation is useful for ordinal data and non-linear monotonic relationships.

Quick revision · 6

- Spearman's correlation ranges from -1 to $+1$.
- No tied ranks → use the basic Spearman formula.
- Tied ranks → use average ranks and correction terms.
- Kendall's coefficient of concordance → measures agreement among multiple rankings.
- Correlation matrix → table of pairwise correlation coefficients.

Quick revision · 7

- Diagonal elements of a correlation matrix → 1.
- Correlation matrix → symmetrical about the diagonal.
- Correlogram → graphical representation of a correlation matrix.
- Strong positive correlation → coefficient close to $+1$.
- Strong negative correlation → coefficient close to -1.

Quick revision · 8

- Correlation measures association, not necessarily causation.

Important formulae · 1

Covariance:

$$cov(x,y) = \frac{1}{n}\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})$$

Pearson's correlation coefficient:

$$r = \frac{cov(x,y)}{sd(x)sd(y)}$$

Pearson's correlation coefficient using deviations:

$$r = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^{n}(x_i - \bar{x})^2 \sum_{i=1}^{n}(y_i - \bar{y})^2}}$$

Important formulae · 2

Spearman's rank correlation coefficient without tied ranks:

$$\rho = 1 - \frac{6\sum_{i=1}^{n} d_i^2}{n(n^2-1)}$$

where d_i is the difference between the ranks of the ith pair of observations.

Spearman's rank correlation coefficient with tied ranks:

$$\rho = \frac{2S_0 - T_x - T_y - \sum d_i^2}{2\sqrt{(S_0 - T_x)(S_0 - T_y)}}, \quad S_0 = \frac{n(n^2-1)}{12}$$

Correction for tied ranks in X:

$$T_x = \frac{1}{12}\sum_{i=1}^{s} m_i(m_i^2 - 1)$$

Important formulae · 3

Correction for tied ranks in Y:

$$T_y = \frac{1}{12}\sum_{i=1}^{s'} w_i(w_i^2 - 1)$$

Kendall's coefficient of concordance:

$$W = \frac{12\left[\sum_{i=1}^{n} R_i^2 - \frac{\left(\sum_{i=1}^{n} R_i\right)^2}{n}\right]}{k^2 n(n^2-1)}$$

where R_i is the sum of ranks obtained by the ith object and k is the number of judges.

Self-check from the chapter

1. The strength and direction of linear association are measured by Pearson's _____ coefficient. **[Correlation]**
2. A plot of paired observations used to examine association is called a _____ diagram. **[Scatter]**
3. Variables that tend to increase together have _____ correlation. **[Positive]**
4. Variables that tend to move in opposite directions have _____ correlation. **[Negative]**

5-MINUTE REVISION

Regression Analysis

From the chapter's own revision points, formulae and questions.

Selected from Chapter 9 · Chapter Summary in the textbook.

Textbook authors: **Dr. Pratheesh P. Gopinath · Dr. Adarsh V.S. · Mohammed Hisham M. · Dr. Manju Mary Paul**

STAT 3202 · Chapter revision

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Quick revision · 1

- Regression → models the relationship between a dependent variable and one or more independent variables.
- Dependent variable → response or outcome to be predicted or explained.
- Independent variable → predictor or explanatory variable.
- Simple linear regression → one dependent variable and one independent variable.
- Multiple linear regression → one dependent variable and two or more independent variables.

Quick revision · 2

- Nonlinear regression → mean function nonlinear in its unknown parameters; polynomial regression can be linear in its coefficients.
- Logistic regression → categorical dependent variable.
- Line of best fit → fitted line obtained using least squares.
- Error → theoretical difference from the true regression relationship.
- Residual → observed value minus fitted value.

Quick revision · 3

- Least squares → minimizes the sum of squared residuals.
- Slope β → expected change in y for a one-unit increase in x.
- Intercept α → value of y when $x = 0$.
- Regression coefficient can range from $-\infty$ to $+\infty$.
- Positive regression coefficient → direct relationship.

Quick revision · 4

- Negative regression coefficient → inverse relationship.
- Regression of y on x → predict y from x.
- Regression of x on y → predict x from y.
- The regression line passes through $(\bar{x}, \bar{y})$.
- Both regression coefficients have the same sign.

Quick revision · 5

- $\beta_{yx}\beta_{xy} = r^2$.
- If $r = 0$, both regression coefficients are zero.
- If $r = \pm 1$, the two regression lines coincide.
- If one regression coefficient exceeds 1, the other must be less than 1.
- Both regression coefficients cannot exceed 1 simultaneously.

Quick revision · 6

- Regression coefficients are independent of change of origin but not of change of scale.
- Regression coefficients depend on the correlation and the relative standard deviations of the variables.
- Linearity, independence, homoscedasticity, normality of errors for exact inference, absence of exact multicollinearity, absence of error autocorrelation, and correct model specification are important assumptions.
- Regression lines are unique for a given dataset.
- Correlation measures association; regression models the relationship and allows prediction.

Quick revision · 7

- Correlation is symmetric; regression is asymmetric.
- Correlation coefficient is unit-free and lies between -1 and $+1$.
- Regression coefficients have units and are not bounded.
- Regression does not establish causation by itself.

Important formulae · 1

Straight-line equation:

$$Y = \alpha + \beta X$$

Simple linear regression model:

$$y_i = \alpha + \beta x_i + e_i$$

Residual after fitting:

$$\hat{e}_i = y_i - (\hat{\alpha} + \hat{\beta} x_i)$$

Important formulae · 2

Sum of squared errors:

$$E = \sum_{i=1}^{n} e_i^2$$

Sum of squared errors in terms of the regression model:

$$E = \sum_{i=1}^{n} [y_i - (\alpha + \beta x_i)]^2$$

First normal equation:

$$\sum_{i=1}^{n} y_i = n\alpha + \beta \sum_{i=1}^{n} x_i$$

Important formulae · 3

Second normal equation:

$$\sum_{i=1}^{n} x_i y_i = \alpha \sum_{i=1}^{n} x_i + \beta \sum_{i=1}^{n} x_i^2$$

Mean relationship:

$$\bar{y} = \alpha + \beta \bar{x}$$

Intercept:

$$\hat{\alpha} = \bar{y} - \hat{\beta}\bar{x}$$

Important formulae · 4

Regression coefficient of y on x:

$$\hat{\beta}_{yx} = \frac{\sum x_i y_i - \frac{\sum x_i \sum y_i}{n}}{\sum x_i^2 - \frac{(\sum x_i)^2}{n}}$$

Regression coefficient using covariance and variance:

$$\hat{\beta}_{yx} = \frac{cov(x, y)}{var(x)}$$

Regression equation of y on x:

$$\hat{y} = \hat{\alpha} + \hat{\beta}_{yx} x$$

Important formulae · 5

Regression coefficient of x on y:

$$\hat{\beta}_{xy} = \frac{cov(x, y)}{var(y)}$$

Regression equation of x on y:

$$\hat{x} = \hat{\alpha}_1 + \hat{\beta}_{xy} y$$

Relationship between regression coefficients and correlation:

$$\beta_{yx}\beta_{xy} = r_{xy}^2$$

Important formulae · 6

Correlation coefficient from regression coefficients:

$$|r_{xy}| = \sqrt{\beta_{yx}\beta_{xy}}$$

If both regression coefficients are positive:

$$r_{xy} = \sqrt{\beta_{yx}\beta_{xy}}$$

If both regression coefficients are negative:

$$r_{xy} = -\sqrt{\beta_{yx}\beta_{xy}}$$

Important formulae · 7

Regression coefficient in terms of correlation and standard deviations:

$$\beta_{yx} = r\frac{\sigma_y}{\sigma_x}$$

$$\beta_{xy} = r\frac{\sigma_x}{\sigma_y}$$

Change of origin and scale:

$$X = \frac{x - a}{h}, \qquad Y = \frac{y - b}{k}$$

Transformed regression coefficients:

$$\beta_{YX} = \frac{h}{k}\beta_{yx}$$

$$\beta_{XY} = \frac{k}{h}\beta_{xy}$$

Important formulae · 8

Multiple linear regression model:

$$y = \alpha + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_k x_k + e$$

Self-check from the chapter

1. The variable predicted by a regression model is called the _____ variable. **[Dependent]**
2. A regression model with one predictor and a straight-line mean response is called _____ linear regression. **[Simple]**
3. The coefficient representing change in mean response per unit increase in the predictor is the _____. **[Slope]**
4. The predicted response when the predictor is zero is the _____. **[Intercept]**

5-MINUTE REVISION

Probability

From the chapter's own revision points, formulae and questions.

Selected from Chapter 10 · Chapter Summary in the textbook.

Textbook authors: **Dr. Pratheesh P. Gopinath · Dr. Adarsh V.S. · Mohammed Hisham M. · Dr. Manju Mary Paul**

Quick revision · 1

- Random experiment → experiment whose outcome cannot be predicted with certainty.
- Sample space → set of all possible outcomes.
- Sample point → one possible outcome.
- Random variable → numerical value assigned to outcomes.
- Discrete random variable → takes countable values.

Quick revision · 2

- Continuous random variable → takes values within a range.
- Probability → measure of the likelihood of an event.
- Probability always lies between 0 and 1.
- Impossible event → probability 0.
- Certain event → probability 1.

Quick revision · 3

- Event → one or more outcomes of a random experiment.
- Independent events → occurrence of one does not influence the other.
- Dependent events → occurrence of one influences the other.
- Mutually exclusive events → cannot occur together.
- Exhaustive events → together cover the entire sample space.

Quick revision · 4

- Equally likely events → have equal probability.
- Classical probability → based on exhaustive, mutually exclusive, and equally likely outcomes.
- Statistical probability → based on repeated observations.
- Axiomatic probability → based on the three axioms of probability.
- A.N. Kolmogorov → associated with the axiomatic approach.

Quick revision · 5

- Complement → event not occurring.
- $A \cup B$ → A or B or both.
- $A \cap B$ → both A and B.
- A^c → complement of A.
- Conditional probability → probability of an event given that another event has occurred.

Quick revision · 6

- Multiplication law → used for probability of joint occurrence.
- With replacement and independent random selection → successive draws have the same probabilities and are independent.
- Without replacement → successive selections from a finite set are generally dependent.
- Combination → selection without considering order.
- Bayes' theorem → obtains the probability of a cause or hypothesis given observed evidence.

Quick revision · 7

- Prior probability → probability before considering the evidence.
- Posterior probability → updated probability after considering the evidence.
- Gambler's fallacy → belief that previous independent outcomes influence future outcomes.
- In independent trials, past outcomes do not change the probability of the next outcome.

Important formulae · 1

Probability of an event under equally likely outcomes:

$$P(A) = \frac{\text{Number of favourable outcomes}}{\text{Total number of possible outcomes}}$$

Classical probability:

$$P(E) = \frac{m}{n}$$

where m is the number of favourable outcomes and n is the total number of exhaustive, mutually exclusive, and equally likely outcomes.

Statistical probability:

$$P(E) = \lim_{n \to \infty} \frac{m}{n}$$

Important formulae · 2

Probability range:

$$0 \leq P(E) \leq 1$$

Probability of sample space:

$$P(S) = 1$$

Complement rule:

$$P(A^c) = 1 - P(A)$$

Important formulae · 3

Probability of an event and its complement:

$$P(A) + P(A^c) = 1$$

Additive law of probability:

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Additive law for mutually exclusive events:

$$P(A \cup B) = P(A) + P(B)$$

Important formulae · 4

Conditional probability:

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}, \quad P(B) > 0$$

Multiplication law:

$$P(A \cap B) = P(A)P(B \mid A)$$

Multiplication law for independent events:

$$P(A \cap B) = P(A)P(B)$$

Important formulae · 5

Combination:

$$^nC_r = \frac{n!}{r!(n-r)!}$$

Bayes' theorem for two events:

$$P(A \mid B) = \frac{P(B \mid A)P(A)}{P(B)}, \quad P(B) \neq 0$$

Bayes' theorem for a partition of the sample space:

$$P(E_i \mid A) = \frac{P(E_i)P(A \mid E_i)}{\sum_{k=1}^{n} P(E_k)P(A \mid E_k)}$$

Worked example from the chapter

A fair coin has been tossed 99 times, with heads on every toss. Assuming independent tosses, what is the probability of heads on the 100th toss?

Solution

Answer is $\frac{1}{2}$ as each toss is independent of previous toss.

Self-check from the chapter

1. The set of all possible outcomes of a random experiment is called the _____ space. **[Sample]**
2. A subset of the sample space is called an _____. **[Event]**
3. A real-valued function of the outcomes of a random experiment is called a random _____. **[Variable]**
4. Events that cannot occur together are called mutually _____ events. **[Exclusive]**

5-MINUTE REVISION

Probability Distributions

From the chapter's own revision points, formulae and questions.

Selected from Chapter 11 · Chapter Summary in the textbook.

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Quick revision · 1

- Probability distribution → gives possible values of a random variable and their probabilities.
- Discrete probability distribution → random variable takes countable values.
- Continuous probability distribution → random variable takes continuous values.
- Expected value → long-run average value of a random variable.
- Probability mass function → used for discrete random variables.

Quick revision · 2

- Probability density function → used for continuous random variables.
- For a p.m.f., $p(x) \geq 0$.
- For a p.m.f., $\sum p(x) = 1$.
- Bernoulli distribution → exactly two possible outcomes, success and failure.
- Bernoulli parameter → p, the probability of success.

Quick revision · 3

- Bernoulli mean → p.
- Bernoulli variance → $p(1-p)$.
- Binomial distribution → repeated Bernoulli trials.
- Binomial requirements → fixed number of trials, independent trials, and constant probability of success.
- Binomial parameters → n and p.

Quick revision · 4

- Binomial failure probability → $q = 1 - p$.
- Binomial mean → np.
- Binomial variance → npq.
- In fitting a binomial distribution, $p = \frac{\bar{x}}{n}$.
- Expected frequency → $N \times p(x)$.

Quick revision · 5

- Poisson distribution → models counts under appropriate rate and independence assumptions; it also approximates rare binomial successes.
- Poisson parameter → λ.
- Poisson mean → λ.
- Poisson variance → λ.
- In fitting a Poisson distribution, $\lambda = \bar{x}$.

Quick revision · 6

- Normal distribution → continuous, bell-shaped and symmetric.
- Normal distribution parameters → μ and σ.
- Mean = median = mode in a normal distribution.
- Normal distribution is symmetric about its mean.
- Total area under the normal curve → 1.

Quick revision · 7

- Standard normal distribution → mean 0 and standard deviation 1.
- Z-score → number of standard deviations an observation is from the mean.
- Standardization → subtract the mean and divide by the standard deviation.
- About 68.26% of observations lie within $\mu \pm \sigma$.
- About 95% of observations lie within $\mu \pm 1.96\sigma$.

Quick revision · 8

- About 95.44% of observations lie within $\mu \pm 2\sigma$.
- About 99% of observations lie within $\mu \pm 2.58\sigma$.
- About 99.73% of observations lie within $\mu \pm 3\sigma$.
- Normal distribution has skewness 0.
- Normal distribution has kurtosis 3.

Quick revision · 9

- All odd-order central moments are zero for the normal distribution.
- The total area under the normal curve represents total probability.
- The tails of the normal distribution are asymptotic to the horizontal axis.
- A linear combination of independent normal variables is also normally distributed.
- The Central Limit Theorem is closely associated with the normal distribution.

Quick revision · 10

- Z-table gives cumulative probability to the left of a specified Z-score.
- Probability above a Z-score is obtained by subtracting the cumulative probability from 1.
- Probability between two Z-scores is obtained by subtracting the smaller cumulative probability from the larger one.

Important formulae · 1

Probability mass function for equally likely elementary outcomes:

$$P(X = x) = \frac{\text{Number of outcomes giving } X = x}{\text{Total number of elementary outcomes}}$$

Expected value of a discrete random variable:

$$E(X) = \sum_{x} xp(x)$$

Expected value of a continuous random variable:

$$E(X) = \int_{-\infty}^{\infty} xf(x)\,dx$$

Important formulae · 2

Probability mass function:

$$p(x) = P(X = x)$$

Bernoulli probability mass function:

$$p(x) = p^x(1-p)^{1-x}, \quad x = 0, 1$$

Bernoulli mean:

$$E(X) = p$$

Important formulae · 3

Bernoulli variance:

$$Var(X) = p(1-p)$$

Binomial probability mass function:

$$p(x) = \binom{n}{x} p^x q^{n-x}, \quad x = 0, 1, 2, \ldots, n$$

Combination form of binomial probability:

$$p(x) = \frac{n!}{x!(n-x)!} p^x q^{n-x}$$

Important formulae · 4

Probability of failure in binomial distribution:

$$q = 1 - p$$

Binomial mean:

$$E(X) = np$$

Binomial variance:

$$Var(X) = npq$$

Important formulae · 5

Estimation of binomial probability of success:

$$p = \frac{\bar{x}}{n}$$

Expected frequency in fitting a binomial distribution:

$$\text{Expected frequency} = N \times p(x)$$

Poisson probability mass function:

$$p(x) = \frac{e^{-\lambda}\lambda^x}{x!}, \quad x = 0, 1, 2, \ldots$$

Important formulae · 6

Poisson mean:

$$E(X) = \lambda$$

Poisson variance:

$$Var(X) = \lambda$$

Estimation of Poisson parameter:

$$\lambda = \bar{x}$$

Important formulae · 7

Expected frequency in fitting a Poisson distribution:

$$\text{Expected frequency} = N \times p(x)$$

Normal probability density function:

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}, \quad -\infty < x < +\infty$$

Standard normal transformation:

$$z = \frac{X - \mu}{\sigma}$$

Important formulae · 8

Normal distribution notation:

$$X \sim N(\mu, \sigma^2)$$

Standard normal distribution notation:

$$X \sim N(0, 1)$$

Quartile deviation for a normal distribution:

$$Q.D. \approx 0.67449\sigma$$

Important formulae · 9

Mean deviation for a normal distribution:

$$M.D. = \sigma\sqrt{2/\pi} \approx 0.79788\sigma$$

Worked example from the chapter

What is the z-score of a value of 27, given a set mean of 24, and a standard deviation of 2?

Solution

To find the z-score we need to divide the difference between the value, 27, and the mean, 24, by the standard deviation of the set, 2. $z = \frac{27-24}{2} = \frac{3}{2} = 1.5$. This indicates that 27 is +1.5 standard deviations above the mean.

Self-check from the chapter

1. The probability function of a discrete random variable is called a probability _____ function. **[Mass]**
2. The sum of all probabilities in a discrete distribution is _____. **[One]**
3. The long-run average value of a random variable is its _____ value. **[Expected]**
4. A single trial with outcomes coded 0 and 1 follows a _____ distribution. **[Bernoulli]**

5-MINUTE REVISION

Sample Survey

From the chapter's own revision points, formulae and questions.

Selected from Chapter 12 · Chapter Summary in the textbook.

Textbook authors: **Dr. Pratheesh P. Gopinath · Dr. Adarsh V.S. · Mohammed Hisham M. · Dr. Manju Mary Paul**

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Quick revision · 1

- Sample survey → systematic collection of information from a sample.
- Population → complete collection of units under study.
- Population size → N.
- Sample → subset of the population selected for study.
- Sample size → n.

Quick revision · 2

- Sampling fraction → $\frac{n}{N}$.
- Sampling unit → basic unit selected from the population.
- Census → complete enumeration of every population unit.
- Sampling design → procedure used to select the sample.
- Sampling frame → complete list or source identifying the sampling units.

Quick revision · 3

- Parameter → numerical characteristic of the population.
- Statistic → numerical measure calculated from sample data.
- Estimator → statistic used to estimate a population parameter.
- Estimate → particular value obtained from an estimator.
- Unbiased estimator → estimator whose expected value equals the true parameter.

Quick revision · 4

- Sampling error → difference caused by observing a sample rather than the entire population.
- Sampling error is absent in a census.
- Increasing sample size generally decreases standard error for a fixed design and population.
- Non-sampling error → errors other than sampling error.
- Census data can still contain non-sampling errors.

Quick revision · 5

- Confidence interval → interval estimate of a population parameter.
- Confidence level → $100(1-\alpha)\%$.
- $\alpha = 0.05$ → 95% confidence level.
- $\alpha = 0.01$ → 99% confidence level.
- A 95% confidence interval describes the long-run performance of the interval construction method.

Quick revision · 6

- Probability sampling → selection based on probability laws.
- Simple random sampling → equal inclusion probabilities; successive draws are independent only with replacement.
- SRSWR → selected unit is returned before the next selection.
- SRSWOR → selected unit is not returned.
- SRS requires a complete sampling frame.

Quick revision · 7

- Stratified sampling → population divided into non-overlapping strata and samples selected independently from each stratum.
- Stratification generally improves precision when strata are internally homogeneous.
- Systematic sampling → select a random start and then every kth unit.
- Systematic sampling interval → $k = \frac{N}{n}$.
- Cluster sampling → select clusters randomly and enumerate all units within selected clusters.

Quick revision · 8

- Cluster sampling is useful when individual-unit frames are unavailable and the population is geographically dispersed.
- Probability sampling permits probability-based inference.
- Non-probability sampling → selection based on convenience, judgement, quota, or other non-random criteria.
- Convenience sampling → select easily accessible units.
- Consecutive sampling → collect every accessible eligible unit sequentially until the required number or study period is reached.

Quick revision · 9

- Quota sampling → fill predetermined subgroup quotas using non-random selection.
- Purposive sampling → researcher deliberately selects informative or suitable units.
- Snowball sampling → existing participants refer additional participants.
- Non-probability sampling can be useful when a sampling frame is unavailable or the population is difficult to reach.
- Non-probability sampling generally does not allow sampling error to be measured using probability theory.

Quick revision · 10

- Kiaer → Representative Method, 1895.
- Bowley → introduced random sampling and probability-based inference into survey work.
- P.C. Mahalanobis → pioneered large-scale sample surveys in India.
- Good survey design, representative sampling, trained data collectors, and careful execution are essential for reliable results.

Important formulae · 1

Sampling fraction:

$$f = \frac{n}{N}$$

{#eq-sampfrac}

Systematic sampling interval:

$$k = \frac{N}{n}$$

{#eq-sampinterval}

Confidence level:

$$\text{Confidence level} = 100(1-\alpha)\%$$

{#eq-conflevel}

Important formulae · 2

Expected value of an estimator:

$$E(t)$$

Unbiased estimator:

$$E(t) = \theta$$

{#eq-unbiased}

Expected value of the sample mean:

$$E(\bar{X}) = \mu$$

Self-check from the chapter

1. A numerical constant describing a population is termed a _____. **[Parameter]**
2. A numerical measure calculated from sample observations is termed a _____. **[Statistic]**
3. Complete enumeration of a population is called a _____. **[Census]**
4. The list of sampling units used for sample selection is called the sampling _____. **[Frame]**

5-MINUTE REVISION

Hypothesis Testing

From the chapter's own revision points, formulae and questions.

Selected from Chapter 13 · Chapter Summary in the textbook.

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Quick revision · 1

- Inferential statistics → uses sample data to draw conclusions about a population.
- Hypothesis → tentative and testable statement.
- Statistical hypothesis → testable statement about a population parameter.
- H_0 → null hypothesis; generally represents no effect, no difference, or no relationship.
- H_1 or H_a → alternative hypothesis; represents an effect, difference, or relationship.

Quick revision · 2

- Falsification principle → a scientific hypothesis should be capable of being tested and potentially falsified.
- Karl Popper → associated with the falsification principle.
- Always use "fail to reject H_0"; do not say "accept H_0".
- Test statistic → numerical function of sample observations used to make a statistical decision.
- Common test statistics → Z, t, χ^2, and F.

Quick revision · 3

- Type I error → reject H_0 when H_0 is true.
- Type II error → fail to reject H_0 when H_0 is false.
- α → chosen upper bound on the Type I error probability.
- β → probability of Type II error.
- Power → probability of correctly rejecting a false H_0.

Quick revision · 4

- Power → $1 - \beta$.
- Larger sample sizes generally increase power.
- Sampling distribution → probability distribution of possible values of a test statistic under repeated sampling, assuming H_0 is true.
- Rejection region → values of the test statistic for which H_0 is rejected.
- Critical region → another name for rejection region.

Quick revision · 5

- Acceptance region → complementary region where H_0 is not rejected.
- Critical value → value separating the rejection and non-rejection regions.
- Under H_0, the rejection probability is controlled at α; a discrete test can be conservative.
- Under H_0, the non-rejection probability is at least $1 - \alpha$ for a level-α test.
- $H_1 : \mu < \mu_0$ → left-tailed test.

Quick revision · 6

- $H_1 : \mu > \mu_0$ → right-tailed test.
- $H_1 : \mu \neq \mu_0$ → two-tailed test.
- An equal-tail two-sided test allocates $\alpha/2$ to each tail; a discrete test may not attain these probabilities exactly.
- Right-tailed test → reject H_0 when the calculated statistic is greater than the critical value.
- Left-tailed test → reject H_0 when the calculated statistic is less than the critical value.

Quick revision · 7

- Two-tailed test → reject H_0 when the statistic falls below the lower critical value or above the upper critical value.
- p-value → probability of obtaining a test statistic as extreme as or more extreme than the observed value, assuming H_0 is true.
- Small p-value → strong evidence against H_0.
- Large p-value → insufficient evidence to reject H_0.
- p-value $\leq \alpha$ → reject H_0.

Quick revision · 8

- p-value $> \alpha$ → do not reject H_0.
- Failing to reject H_0 does not prove that H_0 is true.
- The alternative hypothesis determines whether the test is one-tailed or two-tailed.
- The level of significance is selected before conducting the test.
- Common significance levels in the chapter → $\alpha = 0.05$ and $\alpha = 0.01$.

Quick revision · 9

- The eight basic steps are: state H_0, state H_1, set α, collect data, calculate the test statistic, identify the critical value or region, compare, and make the decision.
- A statistical decision can be incorrect because it is based on sample data.
- Type I and Type II errors have different consequences depending on the context.
- Fisher's tea-tasting experiment is a historical illustration of hypothesis testing and significance testing.

Important formulae · 1

Level of significance:

$$P(\text{Reject } H_0 \mid H_0 \text{ is true}) \leq \alpha$$

Probability of Type II error:

$$\beta = P(\text{Fail to reject } H_0 \mid H_0 \text{ is false})$$

Power of a test:

$$\text{Power} = 1 - \beta$$

Important formulae · 2

Size of rejection region:

$$P_{H_0}(\text{Rejection region}) \leq \alpha$$

Size of acceptance region:

$$P_{H_0}(\text{Non-rejection region}) \geq 1 - \alpha$$

Binomial test statistic used in the worked coin example:

$$\hat{p} = \frac{x}{n}$$

Important formulae · 3

Binomial probability:

$$P(X = x) = \binom{n}{x} p^x q^{n-x}$$

Equivalent binomial probability formula:

$$P(X = x) = \frac{n!}{(n-x)!x!} p^x q^{n-x}$$

Decision rule based on p-value:

$$\text{Reject } H_0 \quad \text{if} \quad p\text{-value} \leq \alpha$$

Important formulae · 4

Non-rejection rule based on p-value:

$$\text{Do not reject } H_0 \quad \text{if} \quad p\text{-value} > \alpha$$

Self-check from the chapter

1. A testable statement about a population distribution or its parameters is called a statistical _____. **[Hypothesis]**
2. The hypothesis tested for possible rejection is called the _____ hypothesis. **[Null]**
3. The hypothesis specifying a departure from the null hypothesis is called the _____ hypothesis. **[Alternative]**
4. Rejecting a true null hypothesis is a Type _____ error. **[I]**

5-MINUTE REVISION

Large Sample Tests

From the chapter's own revision points, formulae and questions.

Selected from Chapter 14 · Chapter Summary in the textbook.

Textbook authors: **Dr. Pratheesh P. Gopinath · Dr. Adarsh V.S. · Mohammed Hisham M. · Dr. Manju Mary Paul**

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Quick revision · 1

- Large sample → generally $n \geq 30$.
- Large-sample tests are based on the Central Limit Theorem.
- Central Limit Theorem → sampling distribution of $\bar{X}$ approaches a normal distribution as n increases.
- The observations should be independent.
- The population should have finite variance.

Quick revision · 2

- Large-sample Z-test → based on $N(0,1)$.
- Tests in this chapter → single proportion, equality of two proportions, single mean, equality of two means, and an approximate comparison of two standard deviations under the stated normal-population assumptions.
- Single proportion → compare P with a specified value P_0.
- For a single proportion, $H_0 : P = P_0$.
- For a single proportion, the alternative may be $P < P_0$, $P > P_0$, or $P \neq P_0$.

Quick revision · 3

- Single-proportion standard error uses P_0 and Q_0 under the null hypothesis.
- Two proportions → compare P_1 and P_2.
- For two proportions, $H_0 : P_1 = P_2$.
- The two-proportion test uses the pooled proportion $\widehat{P}$.
- Single mean → compare μ with μ_0.

Quick revision · 4

- When σ is known, use $\sigma/\sqrt{n}$.
- When σ is unknown, use $s/\sqrt{n}$.
- Equality of two means → compare μ_1 and μ_2.
- Equal population standard deviations → use the common σ.
- Equal but unknown standard deviations → use the pooled estimate.

Quick revision · 5

- Unequal population standard deviations → use separate variance terms.
- Two-tailed test → reject if $|Z| > Z_{\alpha/2}$.
- Right-tailed test → reject if $Z > Z_{\alpha}$.
- Left-tailed test → reject if $Z < -Z_{\alpha}$.
- At $\alpha = 0.05$, two-tailed critical value → 1.960.

Quick revision · 6

- At $\alpha = 0.01$, two-tailed critical value → 2.576.
- At $\alpha = 0.05$, right-tailed critical value → 1.645.
- At $\alpha = 0.05$, left-tailed critical value → -1.645.
- The null hypothesis is always stated in terms of the population parameter.
- The test statistic is calculated from sample statistics.

Quick revision · 7

- If the calculated statistic falls in the critical region, reject H_0.
- Otherwise, do not reject H_0.
- For a two-tailed test, α is divided equally between the two tails.
- The sign of Z is important in one-tailed tests; $|Z|$ is used for the two-tailed decision.
- A large absolute value of Z indicates greater departure from the null hypothesis.

Important formulae · 1

Central Limit Theorem:

$$\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \xrightarrow{d} N(0,1)$$

Single population proportion:

$$Z = \frac{p - P_0}{\sqrt{P_0 Q_0 / n}}$$

Complementary proportion:

$$Q_0 = 1 - P_0$$

Important formulae · 2

Pooled proportion for two proportions:

$$\widehat{P} = \frac{n_1 p_1 + n_2 p_2}{n_1 + n_2}$$

Pooled complementary proportion:

$$\widehat{Q} = 1 - \widehat{P}$$

Equality of two proportions:

$$Z = \frac{p_1 - p_2}{\sqrt{\widehat{P}\widehat{Q}\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}$$

Important formulae · 3

Single population mean when σ is known:

$$Z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}}$$

Single population mean when σ is unknown:

$$Z = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$$

Equality of two means when population standard deviations are equal:

$$Z = \frac{\bar{x}_1 - \bar{x}_2}{\sigma\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

Important formulae · 4

Pooled estimate of common standard deviation:

$$\sigma = \sqrt{\frac{n_1 s_1^2 + n_2 s_2^2}{n_1 + n_2}}$$

Equality of two means when population standard deviations are unequal:

$$Z = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

Difference of two standard deviations (population standard deviations unknown):

$$Z = \frac{s_1 - s_2}{\sqrt{\frac{s_1^2}{2n_1} + \frac{s_2^2}{2n_2}}}$$

Self-check from the chapter

1. The theorem supporting the normal approximation to the sample mean under suitable conditions is the _____ Limit Theorem. **[Central]**
2. The standard deviation of a sampling distribution is called the standard _____. **[Error]**
3. The reference distribution used in a Z-test is the standard _____ distribution. **[Normal]**
4. For independent observations with population standard deviation σ, the standard error of the sample mean is _____. **[$\sigma/\sqrt{n}$]**

5-MINUTE REVISION

Small Sample Tests

From the chapter's own revision points, formulae and questions.

Selected from Chapter 15 · Chapter Summary in the textbook.

Textbook authors: **Dr. Pratheesh P. Gopinath · Dr. Adarsh V.S. · Mohammed Hisham M. · Dr. Manju Mary Paul**

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Quick revision · 1

- Small sample → often described as $n < 30$, but this is not a universal boundary.
- Small sample theory → study of sampling distributions for small samples.
- Important small-sample distributions → t, χ^2, and F.
- Student's t-test → used mainly for small samples when σ is unknown.
- Exact one-sample t-test → independent random observations from a normal population with unknown standard deviation.

Quick revision · 2

- One-sample t-test → tests a population mean against a specified value.
- One-sample t degrees of freedom → $n - 1$.
- Two-sample t-test → compares two population means.
- Equal variances → use pooled variance.
- Equal-variance two-sample t degrees of freedom → $n_1 + n_2 - 2$.

Quick revision · 3

- Unequal variances → use Welch's t-test.
- Welch's t-test uses separate sample variance terms.
- Welch's degrees of freedom are generally not an integer and are calculated using the Welch-Satterthwaite formula.
- Paired t-test → compares two related observations.
- Examples of paired data → before and after measurements on the same experimental unit.

Quick revision · 4

- Paired t-test is performed on the differences within pairs.
- Paired t degrees of freedom → $n - 1$.
- Correlation test → tests whether the population correlation coefficient is zero.
- For the correlation test, $H_0 : \rho = 0$.
- Correlation test degrees of freedom → $n - 2$.

Quick revision · 5

- Chi-square tests → based on the chi-square distribution.
- Chi-square tests for categorical counts differ from the parametric variance test for a normal population.
- Important chi-square tests → goodness of fit, independence of attributes, and population variance.
- Goodness-of-fit test → compares observed and expected frequencies.
- Goodness-of-fit null hypothesis → category probabilities follow the specified distribution.

Quick revision · 6

- Goodness-of-fit test is right-tailed.
- Goodness-of-fit degrees of freedom → $n - 1$ when no parameter is estimated.
- Independence test → determines whether two attributes are independent.
- Contingency table → arrangement of frequencies for combinations of categorical attributes.
- Expected frequency → row total × column total ÷ grand total.

Quick revision · 7

- $r \times s$ contingency table degrees of freedom → $(r - 1)(s - 1)$.
- 2×2 contingency table degrees of freedom → 1.
- Yates' correction → optional continuity adjustment for a 2×2 chi-square approximation; assess expected frequencies and consider an exact test for sparse data.
- Population variance test → tests a specified value of σ^2.
- Chi-square variance statistic → $\frac{(n-1)s^2}{\sigma_0^2}$.

Quick revision · 8

- F-test → tests equality of two population variances.
- Placing the larger sample variance first is a convenience for a two-sided F-test with upper-tail area $\alpha/2$.
- With the larger variance in the numerator, $F \geq 1$; with fixed labels, F may be less than 1.
- F degrees of freedom → $n_1 - 1$ and $n_2 - 1$.
- In hypothesis testing, compare the calculated statistic with the appropriate table value.

Quick revision · 9

- For a two-tailed t-test → reject H_0 when $|t|$ exceeds the critical value.
- For a right-tailed t-test → reject H_0 when t exceeds the critical value.
- For a left-tailed t-test → reject H_0 when t is less than the negative critical value.
- Goodness-of-fit and independence tests use the upper chi-square tail; a variance test may be one-sided or two-sided.
- For a two-sided F-test → use both $\alpha/2$ tails, or the equivalent larger-variance ratio with an upper $\alpha/2$ cutoff.

Quick revision · 10

- William Sealy Gosset developed the t distribution and published under the name "Student".
- Student's t distribution has heavier tails than the normal distribution.

Important formulae · 1

Sample variance:

$$s^2 = \frac{\sum_{i=1}^{n}(x_i - \bar{x})^2}{n-1}$$

One-sample t statistic:

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$$

Degrees of freedom for one-sample t-test:

$$df = n - 1$$

Important formulae · 2

Pooled variance:

$$s^2 = \frac{(n_1-1)s_1^2 + (n_2-1)s_2^2}{n_1 + n_2 - 2}$$

Two-sample t statistic for equal variances:

$$t = \frac{\bar{x}_1 - \bar{x}_2}{s\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

Degrees of freedom for equal-variance two-sample t-test:

$$df = n_1 + n_2 - 2$$

Important formulae · 3

Welch t statistic:

$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

Welch degrees of freedom:

$$df = \frac{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)^2}{\frac{s_1^4}{n_1^2(n_1-1)} + \frac{s_2^4}{n_2^2(n_2-1)}}$$

Paired difference:

$$d_i = x_i - y_i$$

Important formulae · 4

Mean paired difference:

$$\bar{d} = \frac{\sum_{i=1}^{n} d_i}{n}$$

Variance of paired differences:

$$s^2 = \frac{\sum_{i=1}^{n}(d_i - \bar{d})^2}{n-1}$$

Paired t statistic:

$$t = \frac{\bar{d}}{s/\sqrt{n}}$$

Important formulae · 5

Degrees of freedom for paired t-test:

$$df = n - 1$$

Test statistic for significance of correlation coefficient:

$$t = \frac{r\sqrt{n-2}}{\sqrt{1-r^2}}$$

Degrees of freedom for correlation test:

$$df = n - 2$$

Important formulae · 6

Chi-square goodness-of-fit statistic:

$$\chi^2 = \sum_{i=1}^{n} \frac{(O_i - E_i)^2}{E_i}$$

Degrees of freedom for goodness-of-fit test:

$$df = n - 1 - m$$

Here n is the number of categories and m is the number of parameters estimated from the data; use $m = 0$ when all probabilities are specified.

Important formulae · 7

Expected frequency in an $r \times s$ contingency table:

$$E_{ij} = \frac{(A_i)(B_j)}{N}$$

Chi-square test of independence:

$$\chi^2 = \sum_{i=1}^{r}\sum_{j=1}^{s} \frac{(O_{ij} - E_{ij})^2}{E_{ij}}$$

Degrees of freedom for an $r \times s$ contingency table:

$$df = (r-1)(s-1)$$

Important formulae · 8

Chi-square statistic for a 2×2 contingency table:

$$\chi^2 = \frac{n(ad-bc)^2}{C_1C_2R_1R_2}$$

Yates' corrected chi-square statistic:

$$\chi^2 = \frac{n[\max(0, |ad-bc| - n/2)]^2}{C_1C_2R_1R_2}$$

Chi-square test statistic for population variance:

$$\chi^2 = \frac{(n-1)s^2}{\sigma_0^2}$$

Important formulae · 9

Degrees of freedom for chi-square variance test:

$$df = n - 1$$

F statistic:

$$F = \frac{s_1^2}{s_2^2}$$

Degrees of freedom for F test:

$$df_1 = n_1 - 1, \qquad df_2 = n_2 - 1$$

Worked example from the chapter

A plant breeder studied the relationship between plant height and grain yield in a sample of 18 rice genotypes and obtained a correlation coefficient of 0.6. Test whether the population correlation coefficient is significant.

Solution

Null hypothesis, H_0: ρ = 0 (population correlation coefficient is zero)

Alternative hypothesis, H_1: $\rho \neq 0$ (population correlation coefficient is not zero)

Sample correlation coefficient (r) = 0.6

Number of pairs (n) = 18

Under the bivariate normal model, the following t statistic is used to test zero correlation:

$$t = \frac{r\sqrt{n-2}}{\sqrt{1-r^2}} = \frac{0.6\sqrt{18-2}}{\sqrt{1-0.36}} = \frac{0.6 \times 4}{\sqrt{0.64}} = \frac{2.4}{0.8} = 3.00$$

t_{cal} = 3.00. The table value of t for $n - 2 = 16$ degrees of freedom at the 5% level of significance (two-tailed) is 2.120. Since the calculated value is greater than the table value, we reject the null hypothesis and conclude that there is a significant correlation between plant height and grain yield.

Self-check from the chapter

1. For a normal population with unknown variance, the standardised sample mean follows Student's _____ distribution. [t]
2. The degrees of freedom of a one-sample t-test are _____. [$n - 1$]
3. The two-sample t-test assuming equal population variances uses a _____ variance estimate. [**Pooled**]
4. The two-sample t-test that does not assume equal population variances is called _____ t-test. [**Welch's**]

5-MINUTE REVISION

Design of Experiments

From the chapter's own revision points, formulae and questions.

Selected from Chapter 16 · Chapter Summary in the textbook.

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STAT 3202 · Chapter revision

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Quick revision · 1

- A well-designed experiment separates the effect of the treatments from other sources of variation, so that observed differences can be attributed to the treatments with confidence.
- Experimental error is the variation among experimental units that receive the same treatment; every design must provide a way of estimating it.
- The three basic principles of design, randomization, replication, and local control, work together: randomization protects against systematic allocation bias, replication permits estimation of experimental error and improves precision, and local control reduces experimental error by accounting for known sources of variation.
- An experimental unit is the smallest unit to which a treatment is independently applied; an observational unit is the unit on which the response is actually measured. Confusing the two leads to pseudo-replication.
- In a Randomized Block Design, similar experimental units are grouped into blocks, and all treatments are randomly assigned within each block, so that treatments are compared under similar conditions.

Quick revision · 2

- Precision can be further improved through good plot technique (keeping everything except the treatment uniform), excluding border plants from data collection (using only the net plot), and appropriate statistical analysis such as ANCOVA, which adjusts for covariates.
- R. A. Fisher developed the statistical foundations of experimental design at Rothamsted Experimental Station in the 1920s and 1930s, introducing ANOVA and, together with Frank Yates, the methodologies that remain central to agricultural research today.

Important concepts · 1

- Experiment → systematic procedure conducted under controlled conditions to discover an effect, test a hypothesis, or illustrate a known effect.
- Design of Experiments → statistical methodology for planning experiments so that treatment effects can be estimated accurately, efficiently, and without bias.
- Main objectives of DoE → validity, reliability, replicability, and optimality.

Important concepts · 2

- Treatment → any object of comparison in an experiment.
- Control → standard treatment used for comparison.
- Experimental unit → smallest unit to which a treatment is independently applied.

Important concepts · 3

- Observation unit → unit on which the response is measured.
- Response → measurable outcome observed after treatment application.
- Factor → variable whose influence on a response is being studied.

Important concepts · 4

- Factor level → value or setting taken by a factor.
- Experimental error → variation in response caused by extraneous factors among experimental units receiving the same treatment.
- Extraneous factors → factors other than the treatments that cause variation in the response.

Important concepts · 5

- Randomization → assignment of treatments to experimental units purely by chance.
- Complete randomization → treatments are assigned without blocking restrictions, subject to the pre-specified treatment counts; allocation may be equal or unequal.
- Replication → application of the same treatment to more than one experimental unit.

Important concepts · 6

- Local control → grouping similar experimental units into blocks before assigning treatments.
- Blocking → another term for local control.
- Basic principles of design → randomization, replication, and local control.

Important concepts · 7

- Randomization → protects against allocation bias and balances unknown sources of variation in expectation; a particular allocation can remain unbalanced.
- Replication → improves reliability and precision and permits estimation of experimental error.
- Local control → reduces experimental error by accounting for known sources of variation.

Important concepts · 8

- Randomized Block Design → design in which similar experimental units are grouped into blocks and all treatments are randomly assigned within each block.
- Border effect → influence of neighbouring plots on plants located near plot borders.
- Net plot → inner portion of a plot from which observations are recorded after excluding border plants.

Important concepts · 9

- Proper plot technique → keeping all management practices other than the treatments as uniform as possible.
- ANCOVA → statistical method that adjusts the response using one or more covariates to improve precision.
- Good experimental design → supports unbiased, precise and interpretable comparisons, uses resources efficiently, and defines the conditions to which conclusions apply.

Self-check from the chapter

1. The condition or combination of conditions applied to an experimental unit is called a _____. **[Treatment]**
2. A standard treatment used for comparison is called a _____. **[Control]**
3. The smallest unit to which a treatment is independently assigned is the _____ unit. **[Experimental]**
4. A variable whose levels are investigated in an experiment is called a _____. **[Factor]**

5-MINUTE REVISION

Uniformity Trials

From the chapter's own revision points, formulae and questions.

Selected from Chapter 17 · Chapter Summary in the textbook.

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Quick revision · 1

- Uniformity trial → used to study the nature and extent of fertility variation in a field.
- A single crop variety is grown over the entire field.
- All management inputs, including fertilizer if used, are applied uniformly.
- A substantial border is removed before dividing the field into basic units.
- Basic unit → smallest unit from which yield is separately recorded.

Quick revision · 2

- Yield differences among basic units → measure of soil heterogeneity.
- Smaller basic units → finer spatial detail, balanced against harvesting accuracy and cost.
- Uniformity trials can help determine → fertility variation, plot size, plot shape, and block size.
- Fertility contour map → graphical representation of soil heterogeneity.
- Moving averages → used to reduce random variation and identify areas of similar fertility.

Quick revision · 3

- Similar fertility areas → grouped together in the contour map.
- Serial correlation → measures the relationship between neighbouring observations.
- Horizontal and vertical serial correlations → used to identify the direction of fertility variation.
- Low serial correlation → little linear association at the chosen lag; this alone does not establish patches.
- High serial correlation → neighbouring yields are associated; use the field map to assess gradients and patches.

Quick revision · 4

- Mean square between strips → simpler method having the same objective as serial correlation.
- Between vertical strips → measures variation across columns, in the horizontal direction.
- Between horizontal strips → measures variation across rows, in the vertical direction.
- Larger strip mean square → indicates greater variation in that direction.
- Fairfield Smith's variance law → relates plot size to the variance of yield per basic unit area.

Quick revision · 5

- Smith's law → $V_x = V_1/x^b$.
- Smith's index of soil heterogeneity → b.
- b → quantitative measure of soil heterogeneity.
- Lower b → slower fitted variance reduction, often associated with positive nearby correlation; a gradient is not established.
- Higher b → a faster fitted reduction in variance as plot size increases; it does not establish a patch pattern.

Quick revision · 6

- Values from 0 to 1 describe the usual idealised range of Smith's index; empirical fitted values can fall outside this range.
- Simulated plots → formed by combining basic units into different sizes and shapes.
- Only plot arrangements that fit exactly within the whole field are used.
- Between-plot sum of squares → calculated for each simulated plot size and shape.
- Variance per basic unit area V_x → sample variance of plot means, equal to $Q_x/(rc - x)$.

Quick revision · 7

- Alternative shapes partition the same field → their variance estimates are dependent.
- Retain shapes separately or state a descriptive averaging rule; an ordinary F test does not justify pooling dependent partitions.
- Smith's index → estimated by regression of $Y = \log V_x - \log V_1$ on $X = \log x$.
- In $Y = cX$, $c = -b$.
- Adjusted b → obtained using interpolation from the adjusted b table.

Quick revision · 8

- Maximum curvature method → used to determine optimum plot size.
- In maximum curvature method → plot size is plotted on the X-axis and CV on the Y-axis.
- As plot size increases → CV generally decreases.
- Apparent bend → a traditional plot-size guide sensitive to axis scales and the available arrangements.
- In the rice example → the field was 12 m × 17 m.

Quick revision · 9

- After removing a 1 m border from all sides → effective area was 10 m × 15 m.
- With 1 m × 1 m basic units → there were 150 basic units.
- In the example, vertical serial correlation → 0.438627.
- In the example, horizontal serial correlation → 0.54317.
- Positive sequence correlations describe nearby similarity; they do not by themselves establish patches or gradients.

Quick revision · 10

- The higher horizontal sequence correlation alone does not establish a horizontal fertility gradient.
- In the example, the larger horizontal-strip mean square indicates greater yield variation across rows; a monotonic gradient is not established.
- For plot size 25 m² in the example → $V_{25} = 1750.14$.
- The calculated Smith's index in the example was $b = 0.3607$ before adjustment.
- The adjusted Smith's index in the example was approximately 0.41156.

Quick revision · 11

- The index summarises variance reduction with plot size; identifying gradients or patches also requires spatial information.
- Uniformity trials are important because field variability exists even when all management practices are kept uniform.

Important formulae · 1

Serial correlation:

$$r_s = \frac{\sum_{i=1}^{n} X_i X_{i+1} - \frac{\left(\sum_{i=1}^{n} X_i\right)^2}{n}}{\sum_{i=1}^{n} X_i^2 - \frac{\left(\sum_{i=1}^{n} X_i\right)^2}{n}}$$

Total number of basic units:

$$n = r \times c$$

Vertical sum of squares:

$$SS_V = \frac{\sum_{i=1}^{c} V_i^2}{r} - \frac{G^2}{n}$$

Important formulae · 2

Horizontal sum of squares:

$$SS_H = \frac{\sum_{i=1}^{r} H_i^2}{c} - \frac{G^2}{n}$$

Vertical mean square:

$$MS_V = \frac{SS_V}{c-1}$$

Horizontal mean square:

$$MS_H = \frac{SS_H}{r-1}$$

Important formulae · 3

Fairfield Smith's variance law:

$$V_x = \frac{V_1}{x^b}$$

Logarithmic form of Smith's variance law:

$$\log V_x = \log V_1 - b \log x$$

Difference form:

$$\log V_x - \log V_1 = -b \log x$$

Important formulae · 4

Regression form:

$$Y = cX$$

where

$$Y = \log V_x - \log V_1$$

and

$$X = \log x$$

Relationship between regression coefficient and Smith's index:

$$c = -b$$

Important formulae · 5

Number of simulated plots:

$$w = \frac{rc}{x}$$

Between-plot sum of squares:

$$Q_x = \sum_{i=1}^{w} \frac{T_i^2}{x} - \frac{G^2}{rc}$$

Variance per unit area:

$$V_x = \frac{Q_x}{rc - x}$$

Important formulae · 6

Coefficient of variation:

$$CV = \frac{\text{Standard deviation}}{\text{Mean}} \times 100$$

Standard deviation from variance:

$$SD = \sqrt{V_x}$$

Adjusted Smith's index:

$$\text{Adjusted } b = y_1 + (b_{\text{cal}} - L_1)\frac{y_2 - y_1}{L_2 - L_1}$$

Important formulae · 7

Descriptive ratio for two plot shapes (not an F test for dependent partitions):

$$R = \frac{\text{Larger between-plot sum of squares}}{\text{Smaller between-plot sum of squares}}$$

Self-check from the chapter

1. A field trial conducted under uniform treatment to study field variability is called a _____ trial. **[Uniformity]**
2. The smallest harvested subdivision in a uniformity trial is called a _____ unit. **[Basic]**
3. A map showing spatial patterns in yield is called a fertility _____ map. **[Contour]**
4. The correlation between yields of neighbouring units is a form of _____ correlation. **[Serial]**

5-MINUTE REVISION

Analysis of Variance

From the chapter's own revision points, formulae and questions.

Selected from Chapter 18 · Chapter Summary in the textbook.

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Quick revision · 1

- ANOVA compares population means by partitioning the total variation.
- The one-way null hypothesis is $H_0 : \mu_1 = \cdots = \mu_v$.
- A mean square is a sum of squares divided by its degrees of freedom.
- The treatment F-ratio compares treatment mean square with error mean square.
- The usual exact F test assumes independent normal errors with a common variance under the specified model.

Quick revision · 2

- A significant overall F indicates that at least one pair of means differs; it does not identify the pairs.
- The additive two-way model separates treatment and block variation.
- With one observation per combination, interaction cannot be estimated separately from error.
- Check the design and residuals before interpreting the F tests.

Important formulae

For a one-way classification with v treatments and n observations, let T_i be the total and n_i the number of observations in treatment i, and G the grand total.

$$CF = \frac{G^2}{n}, \qquad SS_{Total} = \sum_{i,j} y_{ij}^2 - CF$$

$$SS_{Treatment} = \sum_{i=1}^{v} \frac{T_i^2}{n_i} - CF, \qquad SS_E = SS_{Total} - SS_{Treatment}$$

$$MS_{Treatment} = \frac{SS_{Treatment}}{v-1}, \qquad MSE = \frac{SS_E}{n-v}, \qquad F = \frac{MS_{Treatment}}{MSE}$$

For a balanced additive two-way classification with v treatments and r blocks, one observation per combination, and block totals B_j:

$$SS_{Blocks} = \sum_{j=1}^{r} \frac{B_j^2}{v} - CF$$

$$SS_E = SS_{Total} - SS_{Treatment} - SS_{Blocks}, \qquad edf = (v-1)(r-1)$$

Self-check from the chapter

1. ANOVA stands for Analysis of _____. **[Variance]**
2. The null hypothesis in a one-way fixed-effects ANOVA states that all population means are _____. **[Equal]**
3. A sum of squares divided by its degrees of freedom is called a mean _____. **[Square]**
4. The denominator of the usual treatment F-ratio is the _____ mean square. **[Error]**

5-MINUTE REVISION

Multiple Comparison Tests

From the chapter's own revision points, formulae and questions.

Selected from Chapter 19 · Chapter Summary in the textbook.

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Quick revision · 1

- ANOVA → omnibus test for detecting an overall treatment effect.
- Significant ANOVA → indicates that at least one pair of treatment means differs.
- ANOVA alone → does not identify which specific treatment pairs differ.
- Multiple comparison tests → used to identify specific differences among treatment means.
- Protected Fisher's LSD requires a significant ANOVA; Tukey's HSD does not require this preliminary test.

Quick revision · 2

- Type I error → rejecting a true null hypothesis.
- Type II error → failing to reject a false null hypothesis.
- Family → complete set of pairwise comparisons in an experiment.
- FWER → probability of making at least one Type I error within the family of comparisons.
- Number of pairwise comparisons → $v(v-1)/2$.

Quick revision · 3

- Fisher's LSD → two-step procedure involving significant ANOVA followed by pairwise comparisons.
- LSD → Least Significant Difference.
- CD → Critical Difference; commonly used term for LSD in Indian agricultural research.
- Protected LSD → pairwise comparisons are performed only after a significant ANOVA.
- Decision under LSD → if $|\overline{T_i} - \overline{T_j}| > \mathrm{LSD}$, the means differ significantly.

Quick revision · 4

- Equal replications → $\mathrm{LSD} = t_{\alpha/2,edf}\sqrt{2\mathrm{MSE}/r}$.
- LSD is relatively liberal compared with Tukey's test.
- Fisher's LSD controls experiment-wise Type I error effectively for two or three treatments under the usual model, but not generally for more than three, as discussed in the chapter.
- DMRT → Duncan's Multiple Range Test.
- DMRT → useful when comparing many treatment means and widely used in plant sciences.

Quick revision · 5

- DMRT procedure → rank means, calculate standard error, obtain r_p, calculate R_p, and compare differences with the appropriate R_p.
- In DMRT, p → number of ordered treatment means spanned by the comparison, including the two means being compared.
- Adjacent means → $p = 2$.
- One mean between the two means → $p = 3$.
- Two means between the two means → $p = 4$.

Quick revision · 6

- DMRT critical range → increases as p increases.
- DMRT → generally more powerful and more liberal than Tukey's test.
- Tukey's HSD → compares all possible pairs while controlling the family-wise error rate.
- Tukey's test → based on the studentised range statistic q.
- HSD → single critical difference for equal replications.

Quick revision · 7

- Decision under Tukey → if $|\overline{T_i} - \overline{T_j}| > \mathrm{HSD}$, the means differ significantly.
- Tukey's test → more conservative than LSD and DMRT.
- More conservative test → less likely to declare a difference significant.
- In the worked example → LSD = 1.10.
- In the worked example → DMRT R_p values range from 1.10 to 1.21.

Quick revision · 8

- In the worked example → Tukey HSD = 1.59.
- Under LSD and DMRT → all treatment pairs are significant.
- Under Tukey → differences of 1.5 are not significant.
- Tukey therefore identifies fewer significant differences than LSD and DMRT in the example.
- Compact letter grouping → convenient presentation of multiple comparison results.

Quick revision · 9

- Same letter → treatments are not significantly different.
- No common letter → treatments are significantly different.
- A treatment can have more than one letter when it belongs to overlapping non-significant groups.
- Letter grouping is also called alphabet notation.
- Line or underscore notation → older method of presenting groups of means.

Quick revision · 10

- Letter and line notation → convey the same statistical information.
- Dunnett's test → useful when every treatment is compared with a single control.
- Bonferroni method → adjusts significance for multiple comparisons.
- Scheffé's test → useful for general contrasts among means.
- SNK → Student-Newman-Keuls multiple comparison procedure.

Important formulae · 1

Number of pairwise comparisons among v treatments:

$$N = \binom{v}{2} = \frac{v(v-1)}{2}$$

Inflated Type I error rate:

$$\text{Inflated } \alpha = 1 - (1-\alpha)^N$$

Standard error of the difference between two treatment means:

$$SE(d) = \sqrt{\text{MSE}\left(\frac{1}{r_i} + \frac{1}{r_j}\right)}$$

Important formulae · 2

Fisher's LSD:

$$\text{LSD} = t_{\frac{\alpha}{2}, edf} \cdot SE(d)$$

LSD for equal replications:

$$\text{LSD} = t_{\frac{\alpha}{2}, edf} \cdot \sqrt{\frac{2\text{MSE}}{r}}$$

Test statistic for pairwise comparison:

$$t = \frac{\overline{T_i} - \overline{T_j}}{SE(d)}$$

Important formulae · 3

Standard error of a treatment mean in DMRT:

$$SE(\overline{Y}) = \sqrt{\frac{\text{MSE}}{r}}$$

Shortest significant range in DMRT:

$$R_p = r_p \cdot SE(\overline{Y})$$

Tukey's honestly significant difference:

$$\text{HSD} = q_{\alpha,(v,edf)} \cdot \sqrt{\frac{\text{MSE}}{r}}$$

Self-check from the chapter

1. A test of the overall equality of several treatment means is called an _____ test. **[Omnibus]**
2. A significant treatment F-test indicates that not all treatment means are _____. **[Equal]**
3. The probability of at least one Type I error in a family of tests is called the _____ error rate. **[Family-wise]**
4. The number of pairwise comparisons among v treatment means is _____. **[$v(v-1)/2$]**

5-MINUTE REVISION

Single Factor Experiments

From the chapter's own revision points, formulae and questions.

Selected from Chapter 20 · Chapter Summary in the textbook.

Textbook authors: **Dr. Pratheesh P. Gopinath · Dr. Adarsh V.S. · Mohammed Hisham M. · Dr. Manju Mary Paul**

STAT 3202 · Chapter revision

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Quick revision · 1

- A single factor experiment varies only one factor, the treatments, while all other conditions are kept uniform.
- CRD uses randomization and replication without blocking. It is most useful when units are relatively homogeneous.
- CRD is best suited to laboratory and greenhouse experiments; it is rarely used for field experiments because field plots are usually not homogeneous.
- RCBD adds local control (blocking) to randomization and replication, grouping heterogeneous experimental material into homogeneous blocks along a single known source of variation, most often a fertility gradient.
- In RCBD, number of blocks = number of replications, and number of plots per block = number of treatments.

Quick revision · 2

- LSD controls variation in two perpendicular directions at once, using rows and columns as two separate blocking factors, so the number of rows, columns, and treatments are all equal.
- The statistical model of CRD matches one-way ANOVA; the model of RCBD matches two-way ANOVA; LSD requires a three-way classification (rows, columns, treatments).
- An error-df target such as 12 is a practical planning guide, not a universal requirement. Replication should also reflect the desired power, precision, anticipated variability, and available resources.
- When the overall F test for treatments is significant, Fisher's least significant difference test is used to compare treatment pairs.
- Treatment means that share a common letter or symbol in a grouping table are not significantly different from each other.

Quick revision · 3

- Among CRD, RCBD, and LSD (for the same number of treatments and replications), CRD gives the largest error degrees of freedom, since none of the variation is set aside for blocking.
- Latin squares are often used for a modest number of treatments, for practical reasons rather than a mathematical restriction: too few treatments leave almost no error degrees of freedom, and too many require an impractically large number of homogeneous plots.

Important formulae · 1

Total number of experimental units in an equally replicated CRD:

$$n = rv$$

Error degrees of freedom in an equally replicated CRD:

$$edf = n - v = v(r-1)$$

Error degrees of freedom in RCBD:

$$edf = (v-1)(r-1)$$

Important formulae · 2

Error degrees of freedom in LSD:

$$edf = (v-1)(v-2)$$

Critical difference for CRD with equal replication:

$$C.D. = t_{\left(\frac{\alpha}{2},\, error\ df\right)}\sqrt{\frac{2\,MSE}{r}}$$

Critical difference for CRD with unequal replication:

$$C.D. = t_{\left(\frac{\alpha}{2},\, error\ df\right)}\sqrt{MSE\left(\frac{1}{r_i}+\frac{1}{r_j}\right)}$$

Important formulae · 3

Critical difference for RCBD:

$$CD = t_{\frac{\alpha}{2},edf}\sqrt{\frac{2\,MSE}{r}}$$

Critical difference for LSD:

$$C.D. = t_{\left(\frac{\alpha}{2},\, error\ df\right)}\sqrt{\frac{2\,MSE}{v}}$$

Correction factor for LSD:

$$C.F. = \frac{\left(\sum y_{ijk}\right)^2}{v^2}$$

Self-check from the chapter

1. The full form of CRD is Completely Randomized _____. [Design]
2. CRD is most suitable when experimental units are relatively _____. [Homogeneous]
3. The error degrees of freedom in a CRD with N observations and v treatments are _____. [$N - v$]
4. In an RCBD, each block contains every treatment exactly _____. [Once]

5-MINUTE REVISION

Factorial Experiments

From the chapter's own revision points, formulae and questions.

Selected from Chapter 21 · Chapter Summary in the textbook.

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Quick revision · 1

- Factorial experiment → studies two or more factors simultaneously.
- Treatment combinations → all possible combinations of factor levels.
- Number of treatments → product of the number of levels of all factors.
- Main effect → average change in response due to a change in the level of one factor.
- Simple effect → effect of one factor at a particular level of another factor.

Quick revision · 2

- Interaction → effect of one factor changes according to the level of another factor.
- Parallel population mean profiles → no interaction on the response scale used.
- Non-parallel sample mean profiles → an interaction pattern; an appropriate test is needed to assess statistical significance.
- Symmetrical factorial → all factors have the same number of levels.
- Asymmetrical factorial → factors have unequal numbers of levels.

Quick revision · 3

- Complete factorial → all possible treatment combinations are included.
- Incomplete or fractional factorial → only part of the possible combinations are included.
- 2^n factorial → n factors, each at two levels.
- 3^n factorial → n factors, each at three levels.
- 2^2 → 4 treatment combinations.

Quick revision · 4

- 2^3 → 8 treatment combinations.
- 2^4 → 16 treatment combinations.
- 3^2 → 9 treatment combinations.
- 3^3 → 27 treatment combinations.
- In a 2^n factorial, the treatment degrees of freedom are $2^n - 1$.

Quick revision · 5

- In a 3^n factorial, the treatment degrees of freedom are $3^n - 1$.
- In a 2^2 factorial → A, B, and $A \times B$ each have 1 degree of freedom.
- In a 2^3 factorial → A, B, C, $A \times B$, $A \times C$, $B \times C$, and $A \times B \times C$ each have 1 degree of freedom.
- In a 3^2 factorial → A has 2 df, B has 2 df, and $A \times B$ has 4 df.
- For a three-level factor → 2 df can be divided into linear and quadratic components.

Quick revision · 6

- Yates's method → tabular method for calculating effects and sums of squares in 2^n factorial experiments.
- Fisher's algebraic method → expresses factorial effects and interactions algebraically.
- Confounding → deliberately making an interaction inseparable from block effects to reduce block size.
- Complete confounding → the same interaction is confounded in all replications.
- Partial confounding → different interactions are confounded in different replications.

Quick revision · 7

- Higher-order interactions are generally preferred for confounding because main effects and lower-order interactions are usually more important.
- Major advantage of confounding → smaller and more homogeneous blocks, resulting in improved precision.
- Major disadvantage of confounding → loss of information about the confounded interactions.

Important formulae · 1

Number of treatment combinations in a factorial experiment:

$$N = \prod_{i=1}^{n} s_i$$

{#eq-ntreat}

Number of treatment combinations in a 2^n factorial experiment:

$$N = 2^n$$

{#eq-n2n}

Number of treatment combinations in a 3^n factorial experiment:

$$N = 3^n$$

{#eq-n3n}

Important formulae · 2

Main effect of A in a 2^2 factorial experiment:

$$\text{Main effect of A} = \frac{(A)_{i=1} - (A)_{i=0}}{2r}$$

{#eq-maineffA22}

Sum of squares due to A in a 2^2 factorial experiment:

$$SS_A = \frac{[(A)_{i=1} - (A)_{i=0}]^2}{4r}$$

{#eq-ssA22}

Main effect of B in a 2^2 factorial experiment:

$$\text{Main effect of B} = \frac{(B)_{j=1} - (B)_{j=0}}{2r}$$

{#eq-maineffB22}

Important formulae · 3

Sum of squares due to B in a 2^2 factorial experiment:

$$SS_B = \frac{[(B)_{j=1} - (B)_{j=0}]^2}{4r}$$

{#eq-ssB22}

Interaction effect of $A \times B$ in a 2^2 factorial experiment:

$$\text{Interaction effect of } A \times B = \frac{(AB)_{i+j=0} - (AB)_{i+j=1}}{2r}$$

{#eq-intAB22}

Sum of squares due to $A \times B$:

$$SS_{AB} = \frac{[(AB)_{i+j=0} - (AB)_{i+j=1}]^2}{4r}$$

{#eq-ssAB22}

Important formulae · 4

Critical difference for a significant interaction in a 2^2 factorial experiment:

$$CD = t_{\alpha/2,edf}\sqrt{\frac{2MSE}{r}}$$

{#eq-cd22int}

Main effect of A in a 2^3 factorial experiment:

$$\text{Main effect of A} = \frac{(A)_{i=1} - (A)_{i=0}}{4r}$$

{#eq-maineffA23}

Sum of squares due to A in a 2^3 factorial experiment:

$$SS_A = \frac{[(A)_{i=1} - (A)_{i=0}]^2}{8r}$$

{#eq-ssA23}

Important formulae · 5

Interaction effect of $B \times C$ in a 2^3 factorial experiment:

$$\text{Interaction effect of } B \times C = \frac{(BC)_{j+k=0} - (BC)_{j+k=1}}{4r}$$

{#eq-intBC23}

Sum of squares due to $B \times C$:

$$SS_{BC} = \frac{[(BC)_{j+k=0} - (BC)_{j+k=1}]^2}{8r}$$

{#eq-ssBC23}

Interaction effect of $A \times B \times C$:

$$\text{Interaction effect of } A \times B \times C = \frac{(ABC)_{i+j+k=1} - (ABC)_{i+j+k=0}}{4r}$$

{#eq-intABC}

Important formulae · 6

Sum of squares due to $A \times B \times C$:

$$SS_{ABC} = \frac{[(ABC)_{i+j+k=1} - (ABC)_{i+j+k=0}]^2}{8r}$$

{#eq-ssABC}

Critical difference for comparing means of a two-factor interaction in a 2^3 factorial:

$$CD = t_{\alpha/2,edf}\sqrt{\frac{MSE}{r}}$$

{#eq-cd23twofactor}

Critical difference for comparing means when the three-factor interaction is significant:

$$CD = t_{\alpha/2,edf}\sqrt{\frac{2MSE}{r}}$$

{#eq-cd23threefactor}

Important formulae · 7

Linear comparison for a factor at three levels:

$$A_L = (A)_{i=2} - (A)_{i=0}$$

{#eq-linearcomp}

Quadratic comparison for a factor at three levels:

$$A_Q = (A)_{i=2} + (A)_{i=0} - 2(A)_{i=1}$$

{#eq-quadcomp}

Sum of squares due to A in a 3^2 factorial:

$$SS_A = \frac{(A)_{i=0}^2 + (A)_{i=1}^2 + (A)_{i=2}^2}{3r} - CF$$

{#eq-ssA32}

Important formulae · 8

Sum of squares due to B in a 3^2 factorial:

$$SS_B = \frac{(B)_{j=0}^2 + (B)_{j=1}^2 + (B)_{j=2}^2}{3r} - CF$$

{#eq-ssB32}

Sum of squares due to $A \times B$ in a 3^2 factorial:

$$SS_{AB} = SS_{\text{two-way table}} - (SS_A + SS_B)$$

{#eq-ssAB32}

Critical difference for comparing levels of A in a 3^2 factorial:

$$CD = t_{\alpha/2,edf}\sqrt{\frac{2MSE}{3r}}$$

{#eq-cdA32}

Important formulae · 9

Critical difference for comparing levels of A in a 2×3 factorial:

$$CD_A = t_{\alpha/2,edf}\sqrt{\frac{2MSE}{3r}}$$

{#eq-cdA23asym}

Critical difference for comparing levels of B in a 2×3 factorial:

$$CD_B = t_{\alpha/2,edf}\sqrt{\frac{2MSE}{2r}}$$

{#eq-cdB23asym}

Critical difference for comparing means when $A \times B$ is significant:

$$CD = t_{\alpha/2,edf}\sqrt{\frac{2MSE}{r}}$$

{#eq-cdAB23asym}

Self-check from the chapter

1. An experiment containing all combinations of the levels of its factors is called a _____ factorial experiment. **[Full]**
2. The number of treatment combinations is the _____ of the numbers of levels of the factors. **[Product]**
3. The effect of a factor averaged over the levels of the other factors is called its _____ effect. **[Main]**
4. The effect of one factor at a specified level of another is called a _____ effect. **[Simple]**

5-MINUTE REVISION

Analysis of Covariance

From the chapter's own revision points, formulae and questions.

Selected from Chapter 22 · Chapter Summary in the textbook.

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STAT 3202 · Chapter revision

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Quick revision · 1

- ANCOVA combines treatment comparison with regression adjustment.
- A covariate should be relevant to the response and should not be caused by the treatment being compared.
- The common-slope model assumes the same regression slope across treatments; examine treatment-by-covariate interaction.
- Independent errors, appropriate linearity and common residual variance are required for the stated model; normal errors support the usual exact tests.
- Adjustment compares treatment means at a common covariate value.

Quick revision · 2

- Estimating one covariate slope uses one additional error degree of freedom.
- A useful covariate can improve precision, but improvement is not guaranteed.
- Adjustment does not turn an observational comparison into a randomized experiment.
- Use pair-specific standard errors when covariate means differ.

Important formulae · 1

For the CRD with one covariate and a common within-treatment slope:

$$Y_{ij} = \mu + \tau_i + \beta(X_{ij} - \bar{X}) + e_{ij}, \qquad \hat{\beta} = \frac{E_{xy}}{E_{xx}}$$

Here E_{xx}, E_{xy} and E_{yy} are the within-treatment corrected sums of squares and products defined in the chapter, with $E_{xx} > 0$.

$$E = E_{yy} - \frac{E_{xy}^2}{E_{xx}}, \qquad MSE_{adj} = \frac{E}{n - v - 1}$$

With n observed responses, v treatments and one fitted covariate slope, the CRD error degrees of freedom are $n - v - 1$. For the complete RBD with r blocks, use the within-block-and-treatment quantities and error degrees of freedom $(r-1)(v-1) - 1$.

$$\bar{Y}_{i,adj} = \bar{Y}_i - \hat{\beta}(\bar{X}_i - \bar{X})$$

$$F_{Treatment} = \frac{(E_1 - E)/(v-1)}{MSE_{adj}}$$

Important formulae · 2

Here E_1 is the residual sum of squares from the reduced model without treatment effects, retaining the covariate and, in RBD, the blocks. E is from the full model. The treatment comparison is therefore adjusted for the covariate.

Self-check from the chapter

1. ANCOVA combines analysis of variance with _____. **[Regression]**
2. A measured explanatory variable used for statistical adjustment is called a _____. **[Covariate]**
3. The common-slope ANCOVA model assumes equal regression _____ across treatments. **[Slopes]**
4. A covariate measured before treatment application is called a _____ covariate. **[Pre-treatment]**

5-MINUTE REVISION

Missing Plot Analysis

From the chapter's own revision points, formulae and questions.

Selected from Chapter 23 · Chapter Summary in the textbook.

Textbook authors: **Dr. Pratheesh P. Gopinath · Dr. Adarsh V.S. · Mohammed Hisham M. · Dr. Manju Mary Paul**

STAT 3202 · Chapter revision

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Quick revision · 1

- Record why an observation is missing or unusable.
- A poor response caused by a treatment must not automatically be discarded.
- A substituted value is an estimate, not a recovered observation.
- The CRD can usually be fitted directly with unequal treatment sample sizes.
- A missing plot destroys the complete balance of an RCBD or Latin square.

Quick revision · 2

- With one missing plot, the usual complete-design error degrees of freedom decrease by one, provided the model remains estimable.
- Substitution minimizes error SS; treatment SS and comparison standard errors require the stated adjustments.
- Fitting the model to observed responses is an alternative to substitution.
- Missingness related to an unobserved response can bias the analysis; a calculation alone does not resolve this problem.

Important formulae

For a CRD, analyse the observed responses using their actual treatment sample sizes. If all v treatments remain represented:

$$edf = n_{obs} - v, \qquad SE(\bar{Y}_i - \bar{Y}_j) = \sqrt{MSE\left(\frac{1}{n_i} + \frac{1}{n_j}\right)}$$

For one missing observation in an RCBD with v treatments and r blocks:

$$\hat{x} = \frac{rB + vT - G}{(r-1)(v-1)}, \qquad edf = (r-1)(v-1) - 1$$

Here B and T are the observed totals of the affected block and treatment, and G is the observed grand total.

For one missing observation in a Latin square of order v:

$$\hat{x} = \frac{v(R + C + T) - 2G}{(v-1)(v-2)}, \qquad edf = (v-1)(v-2) - 1$$

Here R, C and T are the observed totals of the affected row, column and treatment. In both substitution formulae, all totals exclude the missing value. Use the bias corrections and comparison formulae in the chapter; do not treat the substituted value as an independent observation.

Self-check from the chapter

1. The classical substitution estimate of a missing plot minimizes the error sum of _____. **[Squares]**
2. Estimating one missing observation reduces the error degrees of freedom by _____. **[One]**
3. An estimated missing value is not an additional independent _____. **[Observation]**
4. In the RBD missing-value formula, totals exclude the missing _____. **[Value]**

5-MINUTE REVISION

Data Transformation

From the chapter's own revision points, formulae and questions.

Selected from Chapter 24 · Chapter Summary in the textbook.

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Quick revision · 1

- Data transformation → used when the assumptions required for valid statistical analysis are not satisfied.
- Main purpose → improve validity, precision, and accuracy of statistical analysis.
- Transformation should not be chosen merely to obtain a desired conclusion.
- ANOVA and mean separation → performed on the transformed data.
- Final means → retransformed to the original scale for interpretation.

Quick revision · 2

- Normality → assess model errors using residual diagnostics after transformation.
- Transformations covered → logarithmic, square root, angular, and the Box-Cox family.
- Logarithmic transformation → suitable when standard deviation is proportional to the mean.
- Log transformation → also useful for multiplicative rather than additive effects.
- Logarithms require positive inputs; shifts need justification, and an alternative model may be preferable.

Quick revision · 3

- Common transformation for zero values → $\log(Y+1)$.
- Positive observations below one can be logged directly; their logarithms are negative.
- Square root transformation → appropriate for Poisson-type count data.
- Examples → insect counts, number of defectives, and counts of seeds affected by disease.
- Poisson distribution → $\sigma^2 = \mu$.

Quick revision · 4

- Empirical square root case → $\sigma^2 = c^2\mu$.
- Zero counts → $\sqrt{Y}$ is defined; the worked example uses $\sqrt{Y+1}$ consistently.
- Retransformation of square root mean → square the value and subtract 1.
- Retransformed square root mean → no greater than the original arithmetic mean; the difference may be substantial.
- Back-transformed means are not generally arithmetic means; any bias correction must match the fitted model.

Quick revision · 5

- Angular transformation → appropriate for percentage data derived from binomial counts.
- Binomial variance → $\sigma^2 = p(1-p)/n$.
- The 30%–70% guideline is approximate; use the binomial model and residual diagnostics to assess the analysis.
- A single proportion outside 30%–70% does not automatically require transformation. If used, apply one transformation consistently.
- Angular transformation → $sin^{-1}\sqrt{p}$.

Quick revision · 6

- For 0% → replace with $1/(4n)$.
- For 100% → replace with $1 - 1/(4n)$.
- Retransformation of angular mean → $100 \times sin^2 X$, using the same angular unit as in the forward transformation.
- Percentage of protein, carbohydrate, etc. → need not necessarily be transformed because such percentages are not necessarily binomial proportions.
- Distribution and transformation:
  - Poisson → $\sqrt{x}$.
  - Empirical $\sigma^2 = c^2\mu$ → $\sqrt{x}$.
  - Binomial → $sin^{-1}\sqrt{p}$.
  - Empirical $\sigma^2 = c^2\mu^2$ → $\log(x)$.

Quick revision · 7

- Logarithmic example → weed counts were transformed using $X = \log Y$.
- Square root example → larval counts containing zero values were transformed using $X = \sqrt{Y+1}$.
- Angular example → insect mortality percentages were transformed using $X = sin^{-1}\sqrt{p}$.
- In all three examples → ANOVA was performed on transformed values.
- Treatment means → compared using the appropriate critical difference after significant treatment F test.

Quick revision · 8

- Final interpretation → based on retransformed means expressed in the original units.

Important formulae · 1

Logarithmic transformation:

$$X = \log Y$$

Logarithmic transformation with a constant:

$$X = \log(Y + k), \quad k > 0$$

Square root transformation:

$$X = \sqrt{Y}$$

Important formulae · 2

Square root transformation when zero values occur:

$$X = \sqrt{Y+1}$$

Alternative transformation for very small counts:

$$X = \sqrt{Y} + \sqrt{Y+1}$$

Angular transformation:

$$X = sin^{-1}\sqrt{p}$$

where

$$p = \frac{Y}{100}$$

Important formulae · 3

Replacement for 0%:

$$p = \frac{1}{4n}$$

Replacement for 100%:

$$p = 1 - \frac{1}{4n}$$

Retransformation after logarithmic transformation:

$$Y = antilog X$$

Important formulae · 4

Retransformation after $X = \sqrt{Y+1}$:

$$Y = X^2 - 1$$

Retransformation after angular transformation:

$$Y = 100 \times sin^2 X$$

Binomial variance:

$$\sigma^2 = \frac{p(1-p)}{n}$$

Important formulae · 5

Poisson variance:

$$\sigma^2 = \mu$$

Empirical variance relationship for square root transformation:

$$\sigma^2 = c^2\mu$$

Empirical variance relationship for logarithmic transformation:

$$\sigma^2 = c^2\mu^2$$

Important formulae · 6

Coefficient of variation:

$$CV = \frac{SD}{Mean} \times 100$$

Critical difference for transformed means in an RBD:

$$CD = t_{\alpha/2,edf}\sqrt{\frac{2MSE}{r}}$$

Self-check from the chapter

1. A change of measurement scale used before analysis is called a data _____. **[Transformation]**
2. Unequal error variances are described as _____. **[Heteroscedasticity]**
3. A transformation commonly used for positive data with multiplicative effects is the _____ transformation. **[Logarithmic]**
4. A traditional variance-stabilising transformation for Poisson counts is the _____ root transformation. **[Square]**